Biology — MDCAT 2026 master notes

Every distinct Biology MCQ in the corpus, as one fact line. **1,829 fact buckets** built from **2,378 parsed questions** across 25 papers and mocks. `×n` marks how many corpus questions ask that fact — anything marked `×3` or more is a near-certain repeat. No explanations by design.

Facts are grouped by sub-topic inside each unit so the notes read as revision material; a few facts sit in a unit because that is where the paper put them, and are tagged `→uN` when they really belong elsewhere. Coverage is auditable via `../Analysis/notes_buckets.json`.

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u1 — Acellular Life (viruses, prions, viroids)

What a virus is

- Virus = non-cellular infectious agent; replicates **only inside** a living host cell.
- A virus cannot replicate on its own — that is the one thing that is *not* true of viruses.
- Virion = the complete, mature, infectious virus particle. ×3
- Capsid = the protein coat surrounding the genome; its subunits are capsomeres.
- Naked virus = non-enveloped virus.
- Envelope is derived from the host membrane, is sunlight-sensitive, and gives the virus a short survival time outside the host. Enveloped viruses are **not** tolerant to antibodies.
- HIV's envelope is made of **glycoprotein**.
- The genetic code is enclosed by a lipid membrane in the **flu virus** (enveloped).
- A virus recognises its specific host by its **viral envelope** (spike proteins).
- Viruses **can** crystallize, mutate and infect bacteria; they cannot respire or excrete.
- All viruses are **parasites** (obligate intracellular).
- "Virus" is Latin for **poison**.
- Ivanowski (Russian) showed a disease agent passed through a fine filter — **viruses**.
- W. M. Stanley prepared a crystalline extract of **tobacco mosaic virus** in 1935.
- An **oncolytic** virus kills cancer cells and boosts immunity.

Shapes and structure

- Adenovirus — polyhedral capsid with **252 capsomeres**.
- Tobacco mosaic virus — **rod-shaped**. ×2
- Polio virus — **spherical**.
- Bacteriophage — tadpole-shaped; the side of the sheath attached to the head is the **collar**. ×6
- **Lysozyme** acts during the **penetration** stage of bacteriophage infection. ×7
- **Prophage** = phage genome integrated into the host DNA.

Genome types

- HIV = **retrovirus**; genome is diploid **single-stranded RNA**. ×3
- **Reverse transcriptase** makes DNA from RNA; the process is **reverse transcription**. ×2
- HIV's replication enzymes are **inside the capsid**, with the RNA.

- HIV **provirus** = double-stranded DNA.
- dsDNA virus — **smallpox**. Herpes is the DNA virus among polio/influenza/herpes/mumps.
- ssDNA virus — **parvovirus** (the mild-rash virus, B19).
- Enveloped RNA virus that spreads in epidemic form — **influenza**.

Prions and viroids

- **Prions** = infectious proteins that cause disease in humans and animals.
- Only a **single cellular gene** encodes the protein that makes up a prion.
- **Viroid** = sub-viral particle (naked RNA) that infects plants.

Viral disease and prevention

- HIV is transmitted through infected blood and hypodermic syringes. ×2
- The **least** common claimed HIV route is a **mosquito bite** (it does not actually transmit HIV).
- HIV in transfusion is prevented by **screening blood for HIV** before transfusion.
- HIV infection runs in **3** stages.
- CD4 is on helper T cells, suppressor T cells and brain (microglial) cells — **MHC II is not a cell**.
- Viral: hepatitis, polio, AIDS, measles, mumps, influenza. **Syphilis and TB are bacterial.** ×3
- Hepatitis is caused by a **virus**.
- Gonorrhoea ← ***Neisseria gonorrhoeae***, spread by sexual contact.
- Measles ← **Paramyxovirus** (morbilli virus).
- Athlete's foot ← **fungus**.
- *Mycoplasma* is **not** a virus (SARS, FIV, SIV are).
- In AIDS, "D" = **Deficiency**; AIDS is caused by a virus. ×2
- **Vaccination** prevents viral infection; measles and mumps are vaccine-preventable. ×2
- Best way to avoid AIDS = **preventive measures**, not vaccination or medication.
- mRNA vaccines (Pfizer, Moderna) **instruct cells to produce the viral spike protein**.
- Sanitizers work on coronavirus through the envelope's **pH sensitivity**.
- Temporary travel restrictions **do** help control viral spread.

Bled in from other units (same buckets)

- Plastids occur only in **plants**. →u4
- Restriction endonucleases = the "molecular scissors" used to cut a gene out of DNA; they occur naturally in **bacteria**. ×3 →u16
- Helper T cells are required to activate B cells. →u12
- Gestation occurs in the **uterus** (pear-shaped, muscular, distensible sac). →u8
- Labour pains come from oxytocin acting on the **myometrium**. →u8

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u2 — Bioenergetics (photosynthesis + cellular respiration)

Light reactions

- Site = **thylakoid** membrane; ATP synthesis during photosynthesis also happens there. ×2
- **Non-cyclic** photophosphorylation is the **Z-scheme**. ×2

- Cyclic photophosphorylation yields **ATP only** (no NADPH, no O_2).
- Cyclic phosphorylation is stimulated by a **low level of NADPH**; C4/cyclic flow gives ATP.
- Non-cyclic flow leaves chlorophyll electron-deficient **twice** (once at PSII, once at PSI).
- PSII gets its replacement electrons from the **photolysis of water**. ×2
- Final acceptor of both electrons **and** protons in the light reactions = **$NADP^+$**.
- **NADP reductase** reduces $NADP^+$ to NADPH.
- End products of the light reactions = **ATP and NADPH**; these are what the dark reaction uses. ×2
- **Plastocyanin** is the only **copper**-containing protein of the light reactions. ×2
- Plastocyanin is reduced by the **cytochrome complex**.
- Without the cytochrome complex, the **proton gradient across the thylakoid membrane** fails.
- When light hits **P700** it is **oxidised** (loses an electron).
- PSI's chlorophyll a absorbs maximally at **700 nm** (PSII at 680 nm).
- Each **photosystem** = light-gathering antenna complex + reaction centre.
- Energy transfer order: **carotenoids → chlorophyll b → chlorophyll a**.
- **Reduction of CO_2 is not** a light-dependent event.
- **Oxygen is not required** for photosynthesis (it is a product).
- **Van Niel** hypothesised that plants split water and release O_2 as a by-product.

Chlorophyll and pigments

- The **porphyrin ring** absorbs the light; **Mg** sits at its centre; the phytol tail anchors it.
- Chlorophyll a carries a **$-CH_3$ on the 2nd pyrrole ring** (chlorophyll b carries –CHO there). ×2
- Chlorophyll a has **72 hydrogen** atoms ($C_{55}H_{72}O_5N_4Mg$) and is **bluish green**. ×2
- **Carotenoids** are the pigments common to plants and humans.
- Light **quality, intensity and duration** all affect photosynthesis — intensity is the usual limiting factor asked for.
- Photosynthesis is maximal in **red** light.

Dark reaction / Calvin cycle

- Chloroplast interior: **grana** embedded in the **stroma** matrix; CO_2 → sugar happens in the **stroma**. ×2
- **Rubisco** is the Calvin-cycle enzyme and the **most abundant protein in nature**. ×3
- CO_2 + **RuBP** → an **unstable 6-carbon intermediate**, which splits into two 3-PGA. ×4
- **RuBP** is the CO_2 acceptor.
- **G3P (glyceraldehyde-3-phosphate)** is the primary product used for glucose synthesis. ×3
- **Regeneration** = 5 G3P → 3 RuBP using ATP. ×6
- The Calvin cycle runs in **3** stages (fixation, reduction, regeneration).
- **Photorespiration** = sugar oxidised in the chloroplast in daylight with no energy yield.
- **Autotrophic** nutrition depends on **photosynthesis**. ×2
- The ultimate energy source of any ecosystem is the **sun**.

Glycolysis

- Site = **cytoplasm**; needs **no oxygen** and runs whether or not O_2 is present. ×2
- Glucose → **2 pyruvate** (3-carbon each). ×3

- Yield = **4 ATP produced, 2 consumed → net 2 ATP** + 2 NADH. ×2
- The **oxidative phase begins with the dehydrogenation of G3P**.
- The isomerisation step interconverts **DHAP ⇌ G3P**.
- Glycerate-1,3-bisphosphate → glycerate-3-phosphate by **losing one phosphate** (to ADP).
- **Kinase** transfers a phosphate from ATP onto glucose (hexokinase step).
- End product of glycolysis under anaerobic conditions = **pyruvate**.

Fermentation / anaerobic respiration

- Anaerobic yield = **2 ATP**; only **2%** of glucose's bond energy reaches ATP. ×6
- Anaerobic respiration gives **less ATP but is faster** than aerobic. ×2
- In humans/muscle the end product is **lactic acid**; in yeast it is **ethyl alcohol + CO_2**.
- **Zymase** (yeast) ferments glucose; alcoholic fermentation = glucose broken down by microorganisms.
- **Anaerobic respiration is used in baking** (and brewing). ×2

Link reaction and Krebs cycle

- Pyruvate → **acetyl CoA** uses **NAD^+ → NADH** and releases CO_2.
- **Acetyl CoA** is the common entry molecule for fat and protein catabolism too.
- Link reaction, Krebs cycle and ETC all occur in the **mitochondrion**. ×2
- Krebs cycle makes **3 NADH per turn → 6 NADH per glucose**. ×2
- **$FADH_2$** is formed at **succinate → fumarate**.
- **Malate** is oxidised by NAD^+ to **oxaloacetate**.
- Krebs cycle's importance = it feeds **oxidative phosphorylation** to make ATP.

Electron transport chain

- Site = **inner mitochondrial membrane / cristae**; ATP synthase sits there. ×3
- **Oxidative phosphorylation** = the phosphorylation when NADH hands electrons to coenzyme Q on the inner mitochondrial membrane. ×7
- Oxidative phosphorylation = ATP synthesis by coupled **chemical + osmotic** events (chemiosmosis) and occurs in **all aerobic cells**.
- First acceptor from **$FADH_2$** = **coenzyme Q**; both NADH and $FADH_2$ feed coenzyme Q. ×6
- Final/ultimate electron acceptor = **oxygen**, which sits at a **low energy level**. ×3
- End product of the ETC = **water**. ×6
- The major event of the ETC is **ATP synthesis**; **34 of the 36 ATP** come from it. ×2
- Aerobic respiration of one glucose = **36 ATP** (38 gross in some texts).
- **Mitochondria are required** for aerobic respiration.
- **Cytochrome c** is the respiratory protein shared across most organisms (evolutionary evidence).
- **Respiration** = biochemical breakdown of complex compounds to release energy.
- **Glyceraldehyde and dihydroxyacetone** are intermediates of both respiration and photosynthesis.

Bled in from other units (same buckets)

- **Spectrophotometry** measures bacterial growth, rate of photosynthesis and minute DNA quantities. →u16
- Polysaccharides are built by **condensation**; **hydrolysis** is its reverse. →u3
- Fat + caustic soda (NaOH) → **saponification**. ×2 →u3

- Ethanol → ethanoic acid by **oxidation**. →u3
- **Fossils** are evidence of past life and of evolutionary history. →u7
- The **Golgi complex** processes and packages cell secretions. →u4

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u3 — Biological Molecules

Water

- Water makes up ≈ **three-fourths of body weight** and is the one compound present in the **same percentage in bacterial and mammalian cells**. ×2
- **Hydrophobic exclusion** is the property of water that keeps lipid membranes intact.
- **Adhesion** = attraction between water and a container / the xylem wall (**cohesion** is water-to-water). ×2
- Ice is less dense because hydrogen bonding makes water **expand** on freezing.
- In the ice lattice each O atom is attached to **4 hydrogen** atoms.

Carbohydrates — general

- Composed of **C, H and O** only, with H : O ≈ **2 : 1**, hence "**hydrated carbons**". ×4
- General formula of a monosaccharide = **$(CH_2O)_n$**.
- Carbohydrates = compounds that on hydrolysis yield **polyhydroxy aldehyde or ketone** subunits.
- **Lactic acid ($C_3H_6O_3$) is not** a carbohydrate.
- **Glucose** is the smallest carbohydrate — it cannot be hydrolysed further.
- **Monosaccharides are the sweetest**; **oligosaccharides** (6–10 units) are less sweet and less soluble; **polysaccharides** have high molecular weight and are sparingly soluble.
- Lactose, maltose and sucrose are all **disaccharides**.
- The four fundamental biomolecules of protoplasm: carbohydrates, proteins, lipids and **nucleic acids**; the most abundant element listed for the human body is **carbon**.

Monosaccharides

- **Glucose** = hexose in the **aldehyde** form; glucose and fructose are both **hexoses**. ×7
- **Erythrose** = a **tetrose**; ribose and deoxyribose have **5** carbons.
- In solution glucose closes into a 6-cornered **glucopyranose** ring, ribose into a 5-cornered **ribofuranose** ring.
- Simplest **keto** monosaccharide = **dihydroxyacetone** (glyceraldehyde is the aldose). ×2
- **Ribose** is the sugar of RNA; **$C_5H_{10}O_5$** is that ribose.
- Normal glucose content of the human body = **0.08%**.
- Most common respiratory substrate = **glucose**; the **direct** energy source in cells = **ATP**.

Disaccharides

- The covalent bridge between two monosaccharides = **glycosidic bond**, formed by **removal of water** (condensation). ×5
- **Maltose** = glucose + glucose, **1→4** glycosidic linkage. ×3
- **Sucrose** = glucose + fructose, **α1→β2** linkage; hydrolysis gives glucose + fructose. ×4

- Sucrose is unreactive/non-reducing because its **functional groups are unavailable**, which is why plants translocate food as **sucrose**; sugarcane stores sucrose. ×2
- **Lactose** = milk sugar = **glucose** + **galactose**. ×4

Polysaccharides

- **Cellulose** is the most abundant carbohydrate in nature, ~**50%** of woody tissue; its pure form is **cotton**; hydrolysis yields the disaccharide **cellobiose**; humans **cannot** digest it.
- **Glycogen** gives a **dark reddish-brown** colour with iodine. ×7
- With **starch**, iodine turns intense **blue-black**.
- Starch = polymer of **α-D-glucose**; **amylose** is unbranched with **1→4** glycosidic links; initial hydrolysis of starch yields **maltose**.
- Starch is the polysaccharide of **grains and seeds**; starch and cellulose are **both polymers of glucose** (starch is stored in plants only).
- **Glycogen** is the animal storage polysaccharide, most abundant in **liver cells**, and is also how **bacteria** store food; its synthesis = **glycogenesis**. ×5
- **Glycogen** is the polysaccharide among sucrose / pentose / lactose.
- **Chitin** is a polysaccharide — the arthropod exoskeleton.
- Deposition of **lignin** kills the cell.
- Bacterial cell wall = **amino acid** + **sugar** (peptidoglycan).

Lipids

- 3 fatty acids + **glycerol** → **triglyceride** + 3 water. ×3
- Fatty acid + glycerol join by an **ester bond**; acylglycerols are esters of **fatty acids and alcohols**. ×4
- **Acylglycerol** is the most abundant, richest energy store in plants and animals.
- **Waxes** = fatty acid + **long-chain alcohol**; they waterproof the leaf cuticle and insect cuticle. ×4
- **Steroids** are the lipids that contain **no fatty acid**; the steroid skeleton is 4 fused rings with **17 carbons**, based on **phenanthrene**.
- **Cholesterol** is the precursor of testosterone, progesterone and oestrogens. ×2
- **Gall stones** are mostly **cholesterol**. ×6
- Saturated fats: commoner in animals, more hydrogen, solid at room temperature — the false claim is that they have "multiple double bonds".
- **Oleic acid** is unsaturated; **stearic acid** (C18 saturated) has the highest melting point; **butyric acid** has **4** carbons.
- Fatty acids carry the **–COOH** group; the **C–H bonds** of lipids are what make them energy-rich.
- Lowest energy per gram among milk / ghee / butter / glucose = **milk**.
- **Tristearin** is a lipid; hydrolysis of a fat gives **glycerol** + **fatty acids**; **triose** + **glycerol** are used to build lipid.
- Phospholipid: the **phosphate head** is the polar/hydrophilic end, the **fatty acid tails** form the hydrophobic zone and sit in the **interior** of the membrane. ×8

Amino acids

- **20** kinds of amino acid occur in cells and build most proteins. ×2
- An amino acid = organic compound with **both an amine and a carboxyl group**; in an **α**-amino acid both are on the **same carbon**; the backbone is **C–C–N**.

- R = **H** → **glycine**; R = **CH_3** → **alanine**. ×5
- **Aspartic acid** is an acidic amino acid; **proline** is the cyclic one.
- At intermediate pH an amino acid is a **zwitterion**, **$–N^+H_3$ … $COO^–$**, also called an **internal salt**; α-amino acids largely exist in this dipolar form. ×4
- At **low pH** it exists as a **cation**; in **basic** solution as the anion **$NH_2–CH_2–COO^–$**.
- When acid is added, the **$COO^–$** group acts as the base.
- **Ninhydrin** is the reagent for identifying amino acids.
- Amino acids the body cannot make = **essential** amino acids. ×3

Proteins

- The **peptide bond** joins **C and N**, formed between the –OH of one amino acid and the H of another's amino group with **release of water**; two amino acids → dipeptide by **condensation**. ×9
- Breaking protein back to amino acids = **hydrolysis**. ×3
- A **protein** has >100 residues or MW >1000 (below that, polypeptide).
- **Primary** = number and sequence of amino acids; **secondary** = the α-helix (heat disrupts this level); **tertiary** = unique 3-D shape of the fully folded chain; **quaternary** = the 2α + 2β assembly of haemoglobin. ×6
- **Fibrous** proteins are long strands/fibrils — e.g. **collagen**, the connective-tissue protein; **globular** proteins are spherical and water-soluble (they are **not** what hair is made of). ×3
- **Haemoglobin** is a **transport protein** that carries O_2 from lungs to every tissue and binds **4** O_2 molecules; **574** amino acids in total, **141** per α-chain. ×5
- Oxygen-carrying capacity = the **ability of haemoglobin to bind oxygen**; as keyed, that binding is catalysed by **carbonic anhydrase**.
- **Sickle-cell anaemia** = single substitution, **glutamic acid → valine at position 6**. ×2
- **Disulphide bridges** occur in proteins and are the bond broken/reformed in hair rebonding. ×2
- **Histones** are rich in **basic** amino acids; heavy-metal ions **destabilise proteins**.
- **Insulin and glucagon** are proteins; **enzymes** are proteins; the **active site** is made of a few (**3–12**) amino acids. ×4
- Proteins are nitrogenous, are tissue building blocks and are polymers of amino acids — they are **not** "hydrated carbons"; the skeleton **$H_2N–CH(R)–COOH$** marks a protein.
- Every reaction in the body has **at least one enzyme** for it.
- **Coenzymes** are derived from **vitamins**; the coenzyme **NAD** is built from **nucleotides**. ×4

Nucleic acids

- **Nucleotide** = pentose sugar + phosphate + nitrogenous base; nucleotides differ only in their **nitrogenous base**. A sugar + base alone = **nucleoside**. ×3
- Nucleotides are joined into a strand by **phosphodiester bonds**, i.e. **C–O–P–O–C**. ×3
- **Purines** (adenine, guanine) are **double-ringed**; pyrimidines are single-ringed. ×3
- **Uracil** is the base absent from DNA, so the **A–U** pair cannot occur in DNA; A pairs with **T**. ×4
- **G–C** pairs stabilise DNA because they make **three hydrogen bonds**.
- One turn of the DNA helix spans **10** base pairs.
- A **chromosome** is **DNA + protein**; the nucleus takes up dye because of its **chromatin**. ×4
- **Bacterial** cells have no histones; prokaryotic DNA is a **single circular double-stranded molecule not bound by a membrane**.

- **RNA** contains **uracil in place of thymine**; the **sugar–phosphate backbone** is what mRNA and DNA share.
- Genetic information is stored as the **genetic code**; **3** nucleotides give **64** codons. ×2
- Central dogma: **DNA → mRNA → protein**; **RNA polymerase** carries out transcription.
- **rRNA** provides the platform for protein synthesis; ribosomes are **RNA + protein**, need **Mg^{2+}** to stay structurally stable, and a cluster on one mRNA is a **polysome**. ×4
- The **anticodon** sits on **tRNA**; 20 different amino acids need at least **20** tRNAs.
- Abundance in the cell, high → low: **rRNA > tRNA > mRNA > DNA**.
- Shared DNA/RNA structure means all organisms are **monophyletic**.
- Monosaccharides are components of **DNA, NADP, ATP and ribulose bisphosphate**.
- A sample with C, H, O, N but **no phosphorus** is a **protein**.

Conjugated molecules and membranes

- A **conjugated molecule** = one class joined to another: **glycoprotein** = protein + carbohydrate, **glycolipid** = lipid + carbohydrate, **lipoprotein** = lipid + protein. A molecule of only carbon and hydrogen is **not** conjugated. ×5
- The lung **surfactant** missing in respiratory distress syndrome is a **lipoprotein**. ×6
- **Nucleoproteins** are the ones that do **not** build membranes.
- **Phospholipids** are the basic structural framework of every membrane; the plasma membrane is chemically **lipid + protein**. ×3
- **Singer and Nicholson**'s fluid-mosaic model = "a sea of lipids in which proteins float".
- **Cholesterol** is the membrane component absent in prokaryotes.
- Surface **glycolipids and glycoproteins** distinguish one cell type from another.
- **Both lipids and proteins** make the membrane selectively permeable.
- Permeability order: **lipids > water/glucose > sodium**. ×2
- Facilitated-diffusion carriers are **proteins**.

Bled in from other units (same buckets)

- **Capsomeres** are made of **protein**; an un-enveloped plus-strand RNA virus is characterised by **lacking an envelope**; phage sequence = **landing → tail contraction → penetration → DNA injection**. →u1
- Light phase: one water molecule → **$2H^+ + 2e^- + ½O_2$**. →u2
- In glycolysis, ATP is **not** directly involved when **1,3-bisphosphoglycerate → 3-phosphoglycerate** (that step *makes* ATP). ×6 →u2
- **Kinase** moves the phosphate from ATP to glucose. →u2
- **Nuclear pores** allow RNA and protein to move between nucleus and cytoplasm. ×6 →u4
- **Golgi complex** processes cell secretions; **lysosomes** do intracellular digestion; glycogen and lipid droplets are **cell inclusions**; the **cell** is the simplest independent unit of life. →u4
- **Tay-Sachs** disease lacks a **lipid-digesting** enzyme. →u4
- **Malonate** is the competitive inhibitor example. →u6
- Insulin **increases membrane permeability** to glucose and converts glucose to **lipids**; **glucagon** turns glycogen → glucose; **adrenaline/noradrenaline** are the amino-acid-derived hormones. →u5
- Blood glucose concentration is **not** under direct nervous control. →u5
- **Pacinian corpuscles** sense deep pressure; **olfactory receptors are chemoreceptors**; a neuron conducts **dendrite → cell body → axon**. →u5

- Mature mammalian RBCs have **no nucleus**; most CO_2 travels as **bicarbonate ions**; **carbon monoxide** kills by binding haemoglobin. →u11
- Antibodies have **4** polypeptide chains — 2 light + 2 heavy joined by **disulphide bridges** — with variable sequences on **both** heavy and light chains. ×3 →u12
- Digestion of **carbohydrate** starts in the oral cavity: **ptyalin/diastase/amylase acts on starch**. ×3 →u14
- **Pepsin**: protein → polypeptides. **Carboxypeptidase** hydrolyses the first peptide bond at the **carboxyl** end. **Lipase** splits lipids into fatty acids. →u14
- Protein absorption products = **amino acids and small polypeptides**; **lacteals** carry fats; **chylomicrons** = fats + proteins; the large intestine absorbs **water and salts**; low blood glucose gives **hunger pangs**. →u14
- **ADH** increases **water** reabsorption; dialysing fluid lacks **nitrogenous waste**; **glucose** passes the glomerulus into Bowman's capsule; **ammonotelic** animals excrete ammonia and **birds** excrete uric acid. →u15
- **Phosphocreatine** is the secondary energy source for muscle contraction. →u9
- DNA is synthesised in the **S** phase of interphase. →u10
- **Reverse transcription** copies **viral RNA** into DNA, giving **cDNA**; **Taq polymerase** is the heat-stable one; **heat** separates DNA strands in PCR; restriction endonucleases cut the **phosphodiester bond**; a gene's position is its **locus**; plants carrying foreign DNA are **transgenic**; cystic-fibrosis liposomes are coated with the **healthy gene**. →u16
- Bacterial waste products = **alcohol, acetic acid and lactic acid**; carnivorous plants gain **nitrates**; developing seeds are rich in **both cytokinins and auxins**; in lichens the **fungus** protects its partner from desiccation.
- Chemistry that landed here: 40 g NaOH + 49 g H_2SO_4 → **6.02×10^{23}** water molecules; 10 g glucose in 100 cm^3 → **0.55 M**; **glucose is not** amorphous; an electron-rich nucleus-seeking species is a **nucleophile**; **super glue** bonds broken jewellery; transition-metal compounds with unpaired electrons are **attracted by a magnet**.

u4 — Cell Structure and Function

Prokaryote vs eukaryote

- Prokaryotes lack a nucleus and **all membrane-bound organelles**; they have a cell wall, DNA and ribosomes.
- The one organelle present in **both** prokaryotes and eukaryotes = **ribosomes** (70S in bacteria, 80S in eukaryotes). ×3
- The only structure a eukaryote has that a prokaryote never has = the **nucleus** / nuclear envelope; absent in prokaryotes too: **chloroplasts**, ER, Golgi, mitochondria, nucleolus.
- **E. coli** is a prokaryote; mitosis is **not** seen in it (or in *Nostoc*, cyanobacteria).
- **Mycoplasma** is devoid of a cell wall; a cell without a wall **bursts in a hypotonic solution**.
- Prokaryotes appeared ~**3.5 billion** years ago; **eukaryotes** first evolved ~**1.5 billion** years ago.

Cell wall

- Bacterial wall = **murein / peptidoglycan** = carbohydrate + protein, unique to bacteria. ×5
- In **Gram-negative** bacteria the peptidoglycan layer is **thin**; **Gram-positive** walls carry **teichoic acids** (absent in Gram-negative) and hold **more lipids** by comparison.
- Fungal wall = **chitin**; plant wall = **cellulose**; **deposition of lignin** makes woody tissue rigid.
- The "heterotrophic, multicellular, no cell wall" description = **Kingdom Animalia**.

- The **secondary wall** sits **between the two primary walls**; the layer laid down at cytokinesis is the **middle lamella**, which also acts as the cementing layer between daughter cells.
- The plant wall is **permeable**; the plasma membrane is **selectively permeable**.
- Bacteria hold their shape through the **cell wall**, and **wall + slime** maintain it.

Plasma membrane

- Lipids make up the **lipid bilayer**; **phospholipids (lecithin)** are the main component, with **cholesterol** essential for mechanical stability — without it the membrane breaks and cells burst. ×3
- **Singer and Nicolson** proposed the fluid-mosaic model: proteins float in a fluid **phospholipid** bilayer, heads out / tails in. ×4
- The **carrier protein** controls membrane fluidity; the **proteins** make it differentially permeable.
- Asymmetry = the hydrophobic tails face inward and the hydrophilic heads outward; the two halves are **not identical**. ×2
- **Membrane proteins** move ions in and out; membrane lipids do **not** do recognition, ligand binding or flexibility — those are protein jobs.
- **Adenylate cyclase** in the membrane acts as an **enzyme**; cartilage nutrition arrives by **diffusion**.
- A cell's "drinking" = **pinocytosis**; engulfing solid particles = **phagocytosis** (a form of endocytosis). The lizard tail-detachment example in the papers is keyed as **autolysis**.
- Passive transport = **facilitated diffusion + osmosis**; **endocytosis** and active transport need **energy**. Water across a selective membrane = **osmosis**.

Organelles — mitochondrion and chloroplast

- The **powerhouse of the cell**, and the only organelle required for **aerobic respiration**, is the **mitochondrion**. ×4
- Mitochondria: **double membrane**, inner folds = **cristae**, small knob-like **F_1 particles = ATP synthetase** on the inner membrane; the respiratory-chain carriers sit on the **inner membrane**; the **matrix** holds the enzymes. ×7
- **Cristae** are the site of **phosphorylation of flavoproteins / redox reactions**.
- Mitochondria have their **own circular DNA and 70S ribosomes**, make their **own proteins**, and are the self-replicating organelle. ×3
- **Mitochondria were first seen as granules in muscle cells**, and they sit in the **mid-piece** of the sperm; they are **absent from mature RBCs**.
- Chloroplasts: membrane-bound, containing **pigments, linear ds-DNA, ribosomes and proteins**; **photophosphorylation** takes place here, and it is the sugar-factory of autotrophic eukaryotes. ×5
- Both chloroplast and mitochondrion **cannot live independently** and share 70S ribosomes.
- Only two organelles are **double-membraned**: mitochondria and chloroplasts.

Lysosomes, peroxisomes, microbodies

- **Lysosomes** = single-membrane spherical sacs full of **hydrolytic enzymes**; they do **intracellular digestion**, **autolysis** and **autophagy** — the one thing they do *not* do is process cell secretions. ×5
- **Tay-Sachs disease** = a malfunctioning **lysosome**. ×4
- Digestive vacuoles and autophagosomes = **secondary lysosomes**.
- **Glyoxysomes** appear only at the **seeding stage of oil-seed plants**; **peroxisomes** convert the glycolate of photorespiration into glycine.

Nucleus

- **Nucleoplasm** = the colloidal mixture of ions and salts in the nucleus; chromosomes are **loosely coiled chromatin** except during division.
- The **nucleolus** synthesises **rRNA** and builds **ribosomes**; it is the structure that **disappears during cell division**. ×3
- **Nuclear pores** are the exchange route between nucleus and cytoplasm. ×7
- The nuclear membrane is a **double membrane**, and its **outer membrane is continuous with the ER**; the **number of pores varies** from cell to cell.
- **Gestations and cell-division facts**: DNA is made in the **S** phase; when chromosomes uncoil and nucleoli reappear at the poles the stage is **telophase**. →u10

Endoplasmic reticulum and Golgi

- **RER** = rough, makes proteins, is where the **enzymes of lysosomes** are synthesised, and is where **proteins and lipids become glycoproteins and glycolipids**.
- **SER** = lipid synthesis and metabolism, **detoxification of drugs and poisons** in liver cells, transmission of impulses in muscle, and it makes **cholesterol and phospholipids** in liver cells. ×4
- Secreted-protein route: **RER → SER → Golgi complex → secretory vesicles**.
- The **Golgi complex** (1888, Camillo Golgi) = processing and packaging, cell secretions, and **synthesis of the plant cell wall**; it is made of **dictyosomes**, is most abundant in **secretory cells**, and makes **neurotransmitters** in the nervous pathway. ×6
- Golgi orientation: the **cis (forming) face** takes up young cisternae, the **trans (maturing) face** sheds the old ones; cisternae break into vesicles at the **convex, maturing face**.
- **Bacterial cells** have no Golgi apparatus.
- Ribosomes are **proteins + RNA**, come in **two subunits**, exist either on RER or free in the **cytoplasm**, and a cluster on one mRNA is a **polysome** whose assembly needs **Mg^{2+}**. ×3
- Ribosomes synthesise **proteins**; a **neuron's** RER + Golgi granule clusters are **Nissl's granules** (groups of ribosomes). ×5

Other organelles and inclusions

- **Vacuoles** are bounded by the **tonoplast**; the large central vacuole is characteristic of **plants**, and rupturing the tonoplast is how a plant cell commits suicide.
- **Plastids** are plant-only: **chloroplasts**, the colourless **leucoplasts** (starch-storing **amyloplasts**), and the pigmented **chromoplasts** that aid seed dispersal and pollination.
- **Peroxisomes**, **cristae**, **ribosomes**, **cytoskeleton**: found in both plant and animal cells, absent in prokaryotes. **Centrioles** mark animal cells; plastids mark plant cells.
- **Mesosomes** are invaginations of the **bacterial cell membrane** and stand in for mitochondria — they initiate **DNA replication** and aid cell division. ×5
- **Cytoplasmic streaming (cyclosis)** drives the active movement of mitochondria.
- **Heterocysts** are found **only in cyanobacteria**.

Cell motility

- **Flagella** originate from **basal bodies** and are made of **microfilaments** (they are definitely motile); **pili** are the thin appendages used in **conjugation** to exchange genetic material and to **increase pathogenicity**.

- A centriole has **nine triplets** of microtubules, with **nine outer pairs plus two central single tubules** in cilia and flagella too; two pairs migrating to opposite poles give **18** triplets.
- Spindle fibres attach to the **centromere**, and animal spindles form from **centrioles**; the chromosome tip = **telomere**. ×3

Cell division and meiosis

- **Crossing over** = exchange of segments between **non-sister chromatids of homologous chromosomes** in meiosis I, occurring in **pachytene**; chiasmata form at the **pairing** stage. ×7
- Its outcome: recombinant chromatids and increased **genetic variation**.
- The **centromere** holds sister chromatids together; **centromeres divide in anaphase II**.
- When genes swap and rejoin, the event is **crossing over** — not autosomal linkage.
- A single DNA-protein strand of a chromosome = **chromatid**; the whole thread = **chromosome**.
- Mitosis handles **wound healing and repair**; **mutation** is the main cause of cancer, and only **mitosis** creates new genetic variation in the asexual sense. ×2
- Nucleus is metabolically most active in **interphase**; ER's flattened sacs = **cisternae**.
- Ultracentrifugation separates **nuclei** → **mitochondria** → **ribosomes** (largest to smallest).
- Chromosome number is **fixed for each species**; chromosomes were first observed by **Walther Flemming**.

Chromosomal abnormalities

- **Down's syndrome** = non-disjunction of the **21st** pair → 47 chromosomes, **trisomy 21**. ×4
- **Klinefelter's** = 44 + **XXY** (fertile male in humans); XXY in **Drosophila** is a sterile male.
- Egg with 24 chromosomes from autosomal non-disjunction = **Klinefelter's syndrome**.
- **Meiosis II of the secondary oocyte proceeds to metaphase**; the ovulated oocyte is arrested in **metaphase II**.
- **Histones** arrange DNA into chromosomes; the **locus** = position of a gene on a chromosome. ×4
- Mendel's **independent assortment** is explained by **chromosome behaviour in meiosis**; genes assort independently only when on **non-homologous chromosomes**.

Bled in from other units (same buckets)

- The intake of O_2-rich air and release of CO_2-rich air = **gaseous exchange**; not a liquid property: **osmotic pressure** (that is a solution property). →u13
- Photosynthesis happens in the **chloroplast**, not in the nucleus. →u2
- Cytoplasm is where glycolysis runs. →u2
- **Phenotype** = the form or appearance of a trait; haploid cells = **sperm**; Mendelian genetics applies to **diploid** organisms; Mendel's seven pea traits sat on **seven chromosomes**; the secretor gene is on chromosome **19**, albinism on **11**; **geographic separation** drives reproductive isolation; evolution is measured by change in **allele frequencies**. →u7, →u10
- **Dry heat** oxidises the bacterial wall; **UV** and **moist heat** are the other sterilisers; a capsule prevents phagocytosis; a bacterial wall is **one layer**; bacteria make **spores** in favourable conditions; **plasmids** carry drug-resistance genes and replicate **autonomously** — the false claim is that the chromosome depends on them.
- **Viruses** are not classified by cell membrane (they have none). →u1
- Neuron at rest = **inside negative** relative to outside; the **nuclear** membrane pore count varies. →u5

- **Apoplast theory** describes water and solute moving through cell walls and intercellular gaps; the symplast route is cytoplasmic.
- A cell with chloroplasts and a large fluid-filled vacuole is a **plant cell**; fungal cells lack chloroplasts.
- Smooth muscle is spindle-shaped, autonomic, non-striated, **one nucleus per cell**, and it does **not** line the heart wall. →u9
- **Phagocytosis** takes up solid food particles; **autophagy** is the cell's self-eating; the rapid carrier-mediated uptake of materials is **facilitated diffusion**.

Leftover definitions and one-offs

- **Endosymbiosis**: flagella may have arisen from an ingested **spirochete**; mitochondria from an aerobic bacterium.
- Organism with **no cell wall at all**: *Mycoplasma* (of the cell types), and of the listed organisms **cyanobacteria** were said to lack one — as opposed to fungi, algae and yeast.
- **Robert Hooke** saw the **cell wall** (cork, 1665) before anyone saw the protoplast.
- **Gram-positive** wall = many layers of peptidoglycan; **Gram-negative** walls hold more **lipid** than Gram-positive ones do.
- Water across a **plant cell wall and intercellular gaps** = apoplast route; symplast = through cytoplasm.
- Features animal and plant cells **share**: cytoplasm, nucleus and mitochondria — **chloroplasts are absent** from animal cells. **Centrioles** mark animal cells; **microtubules** are in both plant and animal cells but not in prokaryotes.
- Cell wall may be **absent in fungi and archaea**; fungal walls = chitin, bacterial = peptidoglycan.
- **Mycobacterium tuberculosis**-type thick walls make bacteria hard to kill; sterilisation of the wall constituents is by **dry heat**.
- Neuron resting state: inside the membrane is **negative** relative to the outside = polarised.
- **Adenylate cyclase** = membrane **enzyme**; cartilage nutrition = **diffusion**; passive processes = **facilitated diffusion + osmosis**.
- Mitochondria and chloroplast ribosomes = **70S**; a eukaryotic cell's ribosomes have **two subunits**.
- **Autophagosomes** originate from the **ER**; **glyoxysomes** at the seeding stage of oil-seed plants; **peroxisomes** make glycine from photorespiratory glycolate.
- The **tonoplast** bounds the vacuole; plant-cell suicide ruptures the **tonoplast**.
- **Phagocytosis** = solid uptake; **pinocytosis** = drinking; **autophagy** = self-eating by lysosomes; **autolysis** = the cell digesting itself.
- Chromosome number is **species-specific**; chromosomes were first observed by **Flemming**; DNA is copied in **S** phase; nucleoli re-form at **telophase**; nuclear activity is highest in **interphase**.
- Cancer's main cause = **mutation**; wound healing = **mitosis**.
- Down's = **trisomy 21 / non-disjunction of pair 21** → 47 chromosomes; **44+XXY = Klinefelter's**; XXY in *Drosophila* = sterile male, in humans = fertile male; a 24-chromosome egg gives **Klinefelter's**.
- Oocyte at ovulation = **metaphase II**; meiosis II of the secondary oocyte stalls at **metaphase**.
- **Mesosomes** = bacterial membrane infoldings acting as the stand-in for mitochondria, aiding **DNA replication**; functionally comparable to mitochondria.
- **Chiasmata** form at pairing in **pachytene**; crossing over swaps **non-sister** chromatids of homologues; **centromeres split in anaphase II**.
- A **chromatid** is one DNA-protein strand of a chromosome; spindles attach at the **centromere**.

- Two corpus lines in this unit are unusable OCR debris rather than questions: a table of contents from the ARK compilation, and a merged fragment whose only recoverable facts are that plant polysaccharides are made by **condensation** and that chloroplasts contain **pigment**.

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u5 — Coordination and Control

Neuron structure

- Impulse path: **sensory neuron** → **inter (associative) neuron** → **motor neuron**; the same as **receptors** → **CNS** → **effectors**. ×4
- **Sensory** neurons run receptor → CNS; **motor** neurons run CNS → muscle. The thing acted on at the motor end is the **effector**.
- The **axon** carries the impulse away from the cell body; **dendrites** receive and relay to the cell body. The axon is directly continuous with the cell body.
- Most organelles are in the **cell body**; sensory neuron cell bodies sit in the **dorsal root ganglion**; one long dendron = **sensory neuron**; most CNS neurons are **multipolar**.
- **Neuroglia** give nutrition, protection and myelination — they do **not** conduct.
- The butterfly-shaped core of the spinal cord is **grey matter** (cell bodies); the outer rim is **white matter**; fibre bundles are a **tract** in the CNS and a **nerve** outside it; a cell-body cluster in the CNS is a **nucleus**, outside it a **ganglion**.
- Spinal-cord white matter is **myelinated axons**; matter = mixed fibre; sensory fibres enter the dorsal horn.

Resting and action potential

- Resting membrane potential ≈ **−70 mV**; outside is positive because of **Na^+**. ×4
- **Na^+** sits ~**10× more outside** than inside; at rest the membrane leaks **K^+** out.
- **Depolarisation = inward Na^+**; **repolarisation = outward K^+**; the peak is about **+50 mV**, so the active potential is roughly **0.05 V**. ×2
- Sufficient depolarisation of the neurolemma fires an **action potential**; a **sub-threshold** stimulus cannot produce a spike.
- An action potential lasts **a few milliseconds** (~1 ms) — never seconds or minutes.
- ATP is needed for the **Na^+/K^+ pump**; ions moving through **gates** is passive.
- Speed depends on **axon volume (diameter)** and on myelination; human myelinated fibres carry ~**100–120 m/s**.
- **Saltatory conduction** is in **myelinated axons**, jumping node to node, ~**100× faster** than unmyelinated; unmyelinated axons conduct **continuously**.
- **Nodes of Ranvier** are gaps in the myelin, formed by **Schwann cells** — they exist in myelinated neurons only, and are what allow saltatory conduction.
- A nerve impulse is an **electrochemical wave** along the fibre.
- Hyperpolarisation would not happen if the **potassium gates were not so leaky**.

Synapse

- Transmitters are made in the **presynaptic neuron**, stored in **synaptic vesicles** within the synaptic knob, and released from the **axon ending**. ×3

- They bind receptors on the **postsynaptic membrane** and change its **permeability to certain ions** — that is what triggers the post-synaptic potential. ×7
- **Ca^{2+}** opens at the terminal to release acetylcholine.
- An excitatory transmitter opens **Na^{+} gates**; **Na^{+} diffusion** depolarises the post-synaptic membrane.
- After transmission the transmitter is **broken down by enzymes** (acetylcholinesterase).
- The CNS's abundant **inhibitory** transmitter is **GABA**; **dopamine** is a neurotransmitter; **adrenaline** is hydrolysed by monoamine oxidase.
- Outside the CNS the transmitter is **acetylcholine**; in the sympathetic system it is **adrenaline**.
- An **electrical synapse** is defined by the **absence of neurotransmitter**.
- Nicotine is **excitatory on acetylcholine receptors** — it raises heart rate, blood pressure and muscular activity.

Receptors and senses

- Pain and tissue damage → **nociceptors**. ×3
- Pressure, touch, stretch, motion and sound → **mechanoreceptors**.
- Taste and smell → **chemoreceptors**; light → **photoreceptors**.
- Skin carries at least **5** different sense-receptors; **Pacinian corpuscles** read deep pressure; **Messelner's** read light touch.
- A reflex arc involves receptors, a sensory neuron, the spinal cord, an inter-neuron and a motor neuron — it does **not** involve the brain.
- A **monosynaptic reflex** uses only a sensory and a motor neuron; the pathway itself is the **reflex arc**.
- Impulse route receptor → effector: **sensory** → **inter** → **motor**.

Brain

- **Forebrain** = cerebrum + hypothalamus + hippocampus + amygdala; the **cerebellum is not** forebrain.
- **Hindbrain** = medulla oblongata + pons + cerebellum — the cerebrum is **not** hindbrain.
- **Cerebrum** = thinking, reasoning, problem-solving, conscious activity (the **cortex** is the seat of consciousness), behaviour, long-term memory.
- **Cerebellum** = balance, posture, coordination, smooth accurate motion; flying birds have a well-developed cerebellum.
- **Medulla** = breathing, heart rate, swallowing; the **respiratory centre** and the central CO_2-chemoreceptors are both here. ×2
- **Hypothalamus** = the homeostatic thermostat; holds the **neurosecretory cells** secreting releasing/inhibiting hormones that control the anterior pituitary; sets the sleep–wake cycle; the infundibulum joins it to the pituitary. ×4
- **Hippocampus** = long-term memory; the one function it shares with the cerebellum is **memory formation**.
- **Amygdala** = pleasure, fear, rage, punishment and sexual arousal. ×8
- **Pons** = the sleep↔wakefulness switch; the **midbrain** is the auditory relay and eye-reflex centre.
- **Thalamus** + **cerebrum** = forebrain; the hemispheres join at the **corpus callosum**.
- Shallow groove = **sulcus**, deep groove = **fissure**, raised area = **gyrus**.
- **Dopamine**-producing cells die in Parkinson's disease.

Spinal cord

- It leaves the cranium through the **foramen magnum**, wrapped in **3 meningeal layers**, with **62 spinal nerves** (31 pairs) branching off.

Homeostasis

- The three parts of a homeostatic loop: **receptor → control centre → effector**.
- The control centre is the **hypothalamus**; long-term control (growth, sexual maturity) is by **hormones**, not neurons.
- The immune system is the one that fights pathogens — not the nervous or endocrine system.
- A homeostatic thermostat sits in the **hypothalamus**.

Pituitary

- The **anterior pituitary** is the **master gland**; it is the one known as "master gland of the body" in the papers (the pituitary overall is the master gland).
- **ADH (vasopressin)** is released from the **posterior pituitary**; **oxytocin** also comes from the posterior lobe. Both are made in the **hypothalamus** and carried down through the infundibulum.
- Anterior-lob e hormones that control other glands are **tropic (trophic)** hormones: FSH, LH, TSH, ACTH, GH, prolactin.
- **FSH** and **LH** are made by the **anterior pituitary**, under **hypothalamic** control; FSH is secreted by the pituitary, not by the ovary.
- Anterior pituitary gonadotrophins = **FSH + LH + prolactin**.

Adrenal gland

- The **adrenal gland** is the gland mainly behind **stress hormones**; its **medulla** makes **epinephrine and nor-epinephrine**, the "fight-or-flight" pair. ×2
- The **cortex** makes corticosteroids (cortisol, aldosterone, androgens).
- **Nor-epinephrine** is the hormone with the greater effect on **peripheral vasoconstriction** and so raises blood pressure. ×7
- **Epinephrine** is the hormone used clinically as a **bronchodilator and heart stimulant**.
- **Aldosterone** raises **Na^+ reabsorption** in the nephron and controls blood sodium. ×2
- Cortex destruction → **Addison's disease**; cortical over-activity → **Cushing's disease**.
- The adrenal medulla's chief product = **epinephrine**.

Pancreas and blood sugar

- **Insulin** = from the **β-cells** of the islets of Langerhans; it lowers blood glucose and is what **diabetic** patients get from **genetically engineered *E. coli***. ×4
- **Glucagon** = from the **α-cells**; it is **antagonistic to insulin** and raises blood glucose. ×4
- **Acromegaly** = over-secretion of **growth hormone (STH)**.
- Long-term control of growth and sexual maturity runs through **hormones**.

Thyroid

- **Thyroxine** carries **4 iodine** atoms and is chemically a **hormone** (it is an amino-acid derivative), classified functionally as a **hormonal protein**-type messenger in the papers.
- **I-131** treats thyroid cancer; **Na-24** is used to **study blood circulation**.

- Thyroid over-activity → **Graves' disease** and **exophthalmia** (both thyroid); low thyroid function with an enlarged gland = **goiter**.
- **Myxoedema** = thyroxine deficiency in adults; **cretinism** = in infants.
- Thyroxine is **not** a steroid; **cortisone/oestrogen** are the steroid hormones.

Oxytocin and prolactin

- **Oxytocin** = labour pains (acting on the **myometrium**) and milk ejection/dilating the mammary milk ducts. ×5
- **Prolactin** is the hormone raised in breastfeeding mothers (it makes the milk; oxytocin releases it).
- Gonadotrophins are lowest at the beginning of the menstrual phase.

Other hormone facts

- **Prostaglandins** are the lipophilic hormones found in every tissue, managing inflammation, blood flow and injury protection; the hypothalamus also regulates them; semen's prostaglandins help sperm transport.
- **Endorphins** = made in the **brain**; they give well-being and inhibit pain.
- **Inhibin** comes from **Sertoli cells** and inhibits FSH; **testosterone** comes from the **Leydig (interstitial)** cells.
- Low parathyroid activity drops blood **Ca^{2+}**.
- **ADH** rises when blood water content falls; its absence causes **diabetes insipidus** (dilute urine in quantity).
- **Vasopressin** is the neurosecretory-cell product.
- Hormones are **chemical messengers**; long-term control is hormonal.
- Water-soluble hormones cross the membrane as: **receptor** → **2nd messenger** → **activation of energy**.
- The **thyroid** is the gland of exophthalmia and Graves' disease.

Bled in from other units (same buckets)

- **Kupffer cells** of the liver do phagocytosis. →u12
- **Insulin** is **not** an enzyme (pepsin, sucrase and catalase are). →u6
- **Gastrin** comes from the **pyloric region** of the stomach. →u14
- **Vestigial organs** = structures reduced over evolution with no apparent function. →u7
- The **thymus** is the gland that produces lymphocytes. →u12
- **Golgi** modification of proteins and lipids, and **Nissl's granules** in a neuron, both belong to u4. →u4
- **Keratin** is the protein with the α-helix **secondary structure**. →u3
- **The thymus** makes lymphocytes; **oxytocin** acts on the myometrium, not the kidney or ovaries.
- Plant items landing here: **auxin** drives apical dominance; *Euglena*'s plant-like feature is its **pyrenoid**. →u8
- The **Casparian strip** blocks water and minerals at the endodermis. →u9
- The reflex arc's link inside the spinal cord is the **associative (inter) neuron**; impulses are secreted from the **axon ends** of the neuron.
- Central chemoreceptors reading blood CO_2 sit in the **medulla**.

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u6 — Enzymes

How enzymes work

- An enzyme is a natural substance that **speeds up** a chemical reaction by **lowering the activation energy** — it does not change the equilibrium, does not alter the reaction's direction, and it is **reused** for the next reaction. ×4
- **Activation energy** = the energy needed to weaken bonds enough that they can break on collision. High activation energy → a **slow** reaction; activation energy is **different for the forward and reverse** reaction.
- The essential structure for enzyme activity is the **tertiary** structure; enzymes are **globular proteins** (collagen is not — it is fibrous). ×2
- Enzymes lower activation energy by **stabilising the transition state**; the active site binds the substrate and is **itself** where catalysis happens. ×3
- They let metabolism run at room temperature because they **greatly increase reaction rates**.
- Enzymes are inactivated by UV / X / αβγ rays because radiation changes their **shape**.
- Enzymes are highly active at **37 °C**; a catalyst lets equilibrium be reached **earlier**.

Active site and models

- The dimple or groove on one side of an enzyme = the **active site**; it binds the substrate and takes part in the catalytic reaction.
- **Lock-and-key model** — proposed by **Emil Fischer in 1890**: the active site is **rigid and specific**, one key fits one lock, enzymes are **non-regulatory** and **cannot change their active site**. ×3
- **Induced-fit model** — proposed by **Koshland in 1959**: the active site is **flexible** and reshapes as the substrate binds; such an enzyme can catalyse a **range of related** substrates and is a **regulatory enzyme**. ×3
- Substrate binding induces a conformational change so the enzyme can perform its **catalytic** function.
- Enzymes that can do one specific reaction are **non-regulatory** and **highly specific**; a regulatory enzyme example is **hexokinase** (not urease, maltase or sucrase).

Temperature and pH

- Beyond the optimum temperature the enzyme **denatures** — irreversibly, so an inactive enzyme's recovery point is the **optimum temperature**, while the **maximum temperature** is where it **starts to denature**.
- Human enzymes work best **37–50 °C**; above 50 °C they **permanently lose activity**.
- Optimum pH: **stomach / pepsin ≈ 2.0** (highly acidic; the single-value question keys **2.5**), **pancreatic lipase ≈ 8.0**, **catalase ≈ 7.0**. ×3
- **pH** is the factor that changes the ionisation state of the active site.

Cofactors and coenzymes

- **Apoenzyme** = the enzyme with its non-protein part missing; **holoenzyme** = enzyme + cofactor.
- A cofactor **covalently / permanently** bound to the protein = a **prosthetic group**. ×4
- A **detachable inorganic ion** cofactor = an **activator** (e.g. **Mg^{2+}**); the non-protein part of an enzyme is a **cofactor**.
- **Coenzymes** make up about **1%** of the whole enzyme and are derived from **vitamins**; **NAD** and **FAD** are coenzymes; **NADP reductase** reduces **$NADP^{+}$**.
- Enzymes and coenzymes in common: both are **reused** and both are **derived from vitamins**.

Inhibitors

- **Competitive inhibitors** are **analogous to the substrate** in shape, bind the **active site**, have **structural similarity to the substrate**, and block it. ×3
- **Non-competitive inhibitors** do **not** resemble the substrate, bind a **regulatory / allosteric site**, and **change the shape of the active site**. ×3
- **Irreversible inhibitors** bind **tightly and permanently**, destroying the globular structure and catalytic activity; **cyanide** forms a **covalent bond** at the active site → **irreversible** inhibitor. ×7
- **Allosteric activators** bind away from the active site and **increase** its activity.
- **Penicillin** = a **competitive inhibitor** of the enzyme bacteria use to build their cell walls. ×2
- **Malonate** is the classic competitive inhibitor against succinate dehydrogenase, which converts succinate into **fumarate**.
- Pesticides **DDT and parathion** are inhibitors of key nervous-system enzymes.
- An **uncompetitive** inhibitor binds only the enzyme–substrate complex.

Enzyme deficiency and one-offs

- **Lysosomes** in eukaryotes hold **hydrolytic enzymes**; heterotrophic digestion also uses **hydrolytic enzymes**.
- **Hyaluronidase** helps the sperm penetrate into the ovum. →u8
- **Pepsinogen** is secreted **inactive** so it does not digest its own cells; trypsin is the classic zymogen.
- Missing **adenosine deaminase** → **severe combined immunodeficiency (SCID)**. →u12
- Enzymes **do not** reverse a reaction's direction, and a catalyst does **not** change equilibrium.

Bled in from other units (same buckets)

- **Collagen** is the protein that is *not* globular. →u3
- **Catalase** in the plant protects against the toxic effect of **alcohol metabolism** (hydrogen peroxide). →u2
- **PEP carboxylase** in C_4 plants lets them avoid photorespiration because PEP carboxylase acts as a **strict carboxylase**. →u2
- **Single-stranded binding proteins (SSBPs)** stabilise the separated DNA strand during replication. →u16
- Chemistry that landed here: propyne → propanone by **boiling with water plus $HgSO_4/H_2SO_4$**; methanol from $CO + H_2$ over **ZnO + CuO**; benzene + 3 H_2 → cyclohexane over nickel with benzene at **1/6** of the original volume; vicinal dihalide → alkyne with **$KOH/NaNH_2$**; acetone + water → **propan-2,2-diol**; SO_3 yield rises by **adding more SO_2**; $N_2 + 3H_2 \rightleftharpoons 2NH_3$ optimum ≈ **450 °C**; benzene halogenation uses **$FeCl_3$**; a catalyst lowers **activation energy**; **DDT/parathion** as inhibitors.
- **Acquired characteristics** (Lamarck) = traits developed by **use or disuse**. →u7

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u7 — Evolution

The theorists

- **Darwin**: evolution by **natural selection**; his core phrase is **descent with modification**; he also believed in the **perceived unity of life**, and *On the Origin of Species* is about **speciation**. ×8
- **Alfred Russel Wallace** reached the same conclusions independently.
- **Lamarck**: **use and disuse of organs** + **inheritance of acquired characteristics**. His giraffe explanation = **stretching of necks over generations**. ×4

- Weismann showed **acquired characters are not inherited** (the mason's children get no strong muscles) — this is Lamarck's idea being rejected.
- Lamarck is remembered **for his rejected but appealing theory of heredity**.
- **Aristotle** is credited with the **unity of life** idea.
- **Herbert Spencer** coined "survival of the fittest" — **Darwin** later used it. ×2
- **Cuvier** explained **catastrophism** = change driven by **sudden natural disasters**.
- **Linnaeus** believed in **special creation**.
- **Hugo de Vries** is the mutation-theory figure (never the inheritance-of-acquired-traits one).

Darwin's mechanism

- Natural selection needs: **overproduction of offspring**, **heritable variation**, **struggle for existence**, and the **fit reproducing more**. ×3
- Natural selection acts directly on the **phenotype** and on **heritable variation**; its raw material is **variation and mutation**.
- Natural selection is **not** regional, is **not** selected by a breeder, does **not** act only on random mutations, and it **does** improve a species' adaptation.
- The primary unit selection acts on is the **individual organism** (the unit of *evolution* is the population).
- **Artificial selection** = humans deliberately applying selection pressure — e.g. **dog breeds**; breeding evidence for evolution = **domestication**.
- **Genetic drift** = random change in allele frequencies, biggest effect in **small populations**.
- **Gene flow** between populations **reduces** their genetic differences.
- Evolution mechanisms: **mutation, gene flow, genetic drift, natural selection** — photosynthesis is not one.
- **Hardy–Weinberg** gives the genotype and allele frequencies at equilibrium. ×2
- **The gene pool** = the total number of genes in a population; allele frequencies are the modern measure of evolution.
- **Speciation** = the formation of a new species, driven by **isolation / reproductive isolation**, and also by the accumulation of small genetic changes over many generations. Isolation **reduces gene flow**.
- **Neo-Darwinism** = Darwin's theory **combined with genetics**, with cases including reproductive isolation.
- **Endosymbiont theory**: mitochondria and chloroplasts were once **free-living bacteria** living inside larger cells.
- **Catastrophism** = sudden natural disasters; **special creation** = life made by a supernatural force; **biogenesis** and **spontaneous generation** are the other old life-origin ideas.

Evidence and comparative anatomy

- **Fossils** are the strongest evidence of evolution, studied as **palaeontology**; they also record **chronology**. ×6
- **Homologous** organs = same origin, **different functions** — evidence of **common ancestry** and **divergent evolution**. ×4
- **Analogous** organs = same function, **different origin/structure** — evidence of **convergent evolution**. ×4
- **Vestigial** organs = functional in the ancestor, reduced or useless now, with no apparent function. ×3
- Classic examples: **whale flippers and human forelimbs = homologous**; **bird wings and insect wings = analogous**; the **human appendix = vestigial**; **baleen whale hind-leg bones = vestigial**.
- Molecular, biochemical and embryological records all support evolution; **digestion** is not evidence.

- Industrial melanism = **natural selection**, first observed in the **peppered moth**, and is the best-known textbook example of selection in action. ×3
- **Antibiotic resistance in bacteria** is also **natural selection**.
- Homologous structures prove **common ancestry**, not the same function alone.
- Serially homologous organs = similar structures repeated along one body axis (like the limbs of a vertebrate).

Bled in from other units (same buckets)

- **Chromosomal abnormality of Turner syndrome = XO** (Klinefelter = XXY). →u10
- **Spermatogenesis** happens in the **seminiferous tubules**. →u8
- Insulin converts glucose into **lipids**. →u5
- Enzyme and substrate react at the **active site**; the term **enzyme** was coined by **Kuhn**. →u6
- **Genetic drift / gene pool / mutation** are evolution mechanisms, not continental drift.
- Vestigial organs have no apparent function today.

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u8 — Reproduction

Male reproductive system — anatomy

- **Spermatogenesis** happens in the **seminiferous tubules**, which also hold the mature spermatozoa. ×4
- The **epididymis** is where sperm are **stored and matured**, and it opens into the **vas deferens**. ×3
- **Vasa efferentia** carry sperm out of the testis; the **vas deferens** connects the epididymis to the ejaculatory duct; the **urethra** is the terminal part of the male duct system.
- **Testes** descended from the abdomen into the **scrotum** before birth; the scrotum keeps them cooler for spermatogenesis.
- Accessory glands: the **seminal vesicle** and the **prostate** (which encircles the urethra) add seminal fluid; the **Cowper's (bulbourethral) gland** neutralises the acidity of urine; the **seminal vesicle** joins the sperm duct at the ejaculatory duct.
- The **urethra** is the organ counted as a urogenital organ in males (not the vas deferens).
- The **testes** do *not* produce **FSH** — FSH comes from the pituitary. They do do spermatogenesis, secrete inhibin and secrete testosterone.

Male hormones

- **Testosterone** is made by the **Leydig (interstitial) cells**, which sit **between** the seminiferous tubules. ×2
- Testosterone drives **sperm production** and **male secondary sexual characteristics**; its *not* a job is inhibiting ICSH — it does not *limit* spermatogenesis.
- **Sertoli cells** make **inhibin**, which inhibits **FSH**. ×2
- **FSH** stimulates spermatogenesis; **LH/ICSH** stimulates the Leydig cells to make testosterone.
- The **prostate** makes a milky alkaline fluid; the **seminal vesicles** supply fructose; **prostaglandins** in semen help sperm transport. →u5

Female reproductive system — anatomy

- **Oogenesis** begins in the **ovary**, where oogonia are produced.
- The **ovary** does *not* make **HCG**; it makes **estrogen**, makes **progesterone** and does ovulate.

- The **oviduct (fallopian tube)** sweeps the released egg, carries the ovum toward the uterus, and is the site of **fertilisation**; its other main job is **transport of the ovum to the uterus**.
- The **uterus** is the small, pear-shaped, muscular, distensible sac where **gestation** happens; it receives, retains and nourishes the fertilised ovum, and its outermost layer is the **perimetrium** (then myometrium, then endometrium). ×2
- The uterus opens to the vagina through the **cervix**, a narrow muscular-sphincter ring; if fertilisation does not happen the **endometrium** is the part that degenerates.
- The female system has ovaries, oviducts and cervix — the **seminiferous tubules** are male.

Oogenesis

- **Primary oocytes** give the **first polar body** in the first meiotic division; the **secondary oocyte** gives the **second** polar body at fertilisation. ×6
- An oogonium → primary oocyte → secondary oocyte → ovum + polar bodies.
- **Follicular development**: primordial → primary → secondary → **Graafian follicle** (mature); most follicles fail by **follicle atresia**, meaning the follicle **degenerates**.
- **FSH** stimulates the **primary follicles** at puberty, and drives the ovary to make **oestrogen**.
- The **LH surge** triggers **ovulation**, and **LH** converts the ruptured Graafian follicle into the **corpus luteum**, a glandular mass that secretes **progesterone**. ×3
- **Progesterone** is the hormone that makes the uterus receptive to implantation, and the one the corpus luteum produces. ×6

The menstrual cycle

- **Ovulation** comes **~14 days after the start of menstruation**, so a cycle beginning 10 July ovulates around **24 July**, and the LH surge lands in **days 11–14** of a 28-day cycle. ×7
- The **LH surge** is triggered by the **fall in estrogen / FSH** feedback at mid-cycle.
- Phases, in order: **menstrual** → **follicular (pre-ovulatory, ~9 days, the longest)** → **ovulatory** → **luteal/secretory (~14 days)**.
- The **secretory (luteal) phase** starts after ovulation, has **raised progesterone**, and its endometrium **prepares for implantation**.
- The **proliferative phase** endometrium **regenerates and thickens**.
- Menstruation (endometrial breakdown) is caused by the **fall in progesterone**.
- **Menopause** = the **cessation of ova production** (ovarian degeneration stops).
- Gonadotrophins are **lowest at the start** of the cycle.
- **Oxytocin** causes the labour pains, acting on the **myometrium**, and it also dilates the mammary milk ducts. ×5
- **Prolactin** rises in nursing mothers to make milk.
- Across the placenta, **blood** is *not* exchanged — gases, nutrients and wastages are.

Fertilisation and early development

- **Fertilisation** of the ovum takes place in the **fallopian (oviduct) tube**, at its **proximal part**, in rabbits, humans and other placental mammals. ×2
- **Hyaluronidase** lets the sperm penetrate the ovum. →u6
- **Implantation** of the embryo occurs in the **uterus**, in the endometrium.
- The **corpus callosum** is brain tissue — never a follicle structure.

- **Endosperm** forms through **double fertilisation** in flowering plants.
- Comparative embryology of vertebrates shows development of **gill pouches**; the nervous system develops from the **ectoderm**.
- A bacterium's genome becomes **diploid after binary fission** (once the two copies coexist before segregation).

Sexually transmitted infections

- **AIDS** is the STD that attacks **CD4 (T4) helper lymphocytes** — HIV's target. ×7
- **Syphilis** ← ***Treponema pallidum***, a **spirochete**, incubation **3–4 weeks**. ×5
- **Gonorrhoea** ← ***Neisseria gonorrhoeae***, spread by **unsafe sex** (not vector-borne, not oro-faecal). ×3
- **Vaginal thrush** ← ***Candida albicans***, a yeast of the mucous membrane. ×2
- **Hepatitis B** is the most common STD in the world.
- **AIDS** is sexually transmitted; TB, dengue and cholera are not.

Immunity crossover

- **Artificial active immunity** = a baby receiving **MMR shots**; passive = a fetus or mother receiving the mother's antibodies through the placenta; natural passive = antibodies received in milk; natural active = immunity from a previous infection. →u12

Bled in from other units (same buckets)

- **Fossils** are preserved remains or traces of ancient organisms — evidence for evolution, along with comparative anatomy and comparative embryology; **embryology** is the branch that supplies the embryo-comparison evidence. →u7
- Comparative vertebrate embryology shows **gill pouches**. →u7
- *Rhizopus* zygotes form a dormant thick-walled **zygospore**.
- High **aluminium** may contribute to **Alzheimer's disease**.
- **Menstruation** follows a **fall in progesterone**; the luteinising hormone **triggers ovulation**.
- Chemicals that landed here: **2-methyl-1-pentanol** < **3-methyl-2-pentanol** < **2-methyl-2-pentanol** for nucleophilic C–O bond cleavage reactivity.
- English/LR debris: "rectify the situation"; the larynx's primary function is **voice production** (that belongs to u13); the LH surge falls in **days 11–14**; tissue fluid of the lymphatic system is **lymph** →u11.

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u9 — Support and Movement (skeleton, joints, muscles)

Bone tissue and its cells

- **Osteocytes** = the living cells sitting in the **lacunae** of bone. ×5
- **Chondrocytes** = the living cells of cartilage, sitting in **lacunae** of the cartilage matrix; they are what *form* cartilage. ×8
- **Osteoblasts** = bone-forming cells; **osteoclasts** = bone-resorbing cells; **osteogenic cells** are the stem cells. ×2
- Bone is wrapped by the **periosteum**; cartilage by the **perichondrium**.
- **Collagen** is the protein that establishes the matrix of bone and cartilage; it is most abundant in **tendons**. ×3

- **Hyaline cartilage** is the most abundant cartilage and the type at the ends of long bones; cartilage has **no blood vessels** and heals slowly because it lacks vascular supply.
- Bone is a **connective tissue**; it is ~**70% water** by cell weight (bone tissue itself is mostly mineral).
- The human skeleton = **axial** + **appendicular**.

Skeletal counts

- **22 bones** in the skull; **33 vertebrae** total; **7 cervical** + **12 thoracic** = **19**; the sacrum is the fusion of **5** vertebrae, the coccyx of **4**.
- **30 bones** per half of the upper limb (humerus, radius, ulna + 30 hand bones); the **femur is not** an upper-limb bone and is the **longest bone** of all.
- The pelvic girdle half is **3 bones** (ilium + ischium + pubis fused into the hip bone); the **hyoid** sits between the skull and the postcranial skeleton; the inner ear's **incus, malleus and stapes** are tiny bones.
- Because of the cranium, animals with a skull are **vertebrates**.
- The **temporal** bone is paired (frontal, occipital, vomer and ethmoid are unpaired).
- The first cervical vertebra is the **atlas**, the second the **axis**.
- The brain is protected inside the **cranium**.

Joints

- Joints are studied as **arthrology**; inflammation of a joint = **arthritis**.
- **Synovial joints** have a fluid-filled joint cavity and are the **freely movable** type; the synovial cavity holds the oily fluid that cuts friction.
- **Fibrous joints** are immovable — e.g. the **suture** between skull bones.
- **Cartilaginous joints** are slightly movable — **pubic symphysis**, **intervertebral discs**, costal cartilage; the pubic symphysis is **fibrocartilage**.
- **Hinge joint** = elbow and knee; the **ulna and humerus** meet there, giving back-and-forth movement. Rotational movement at the forearm is **radius and ulna**.
- **Ball-and-socket** = hip and shoulder, the multi-directional joint; the **hip and shoulder** joints are both ball-and-socket (some papers key the same pair as synovial).
- **Gliding (plane) joints** are the slightly movable type.
- **Ligaments** hold bones together; **tendons** connect skeletal muscle to bone.
- **Sciatica** = tingling/pain along the leg; **arthritis** = swollen, painful, immovable joint; **rheumatoid arthritis** is the autoimmune one; **gout** = hard **uric acid crystals** in the joints; **ankylosing spondylitis** fuses the spine; **spondylosis** = degeneration of the intervertebral discs.
- **Rickets** is a vitamin-D/bone-softening disorder; the hardening of cartilage by calcium deposition is **spondylosis**.
- Rib–vertebra attachments are **slightly movable** joints.

Muscle types

- **Skeletal** = voluntary, striated, multinucleate, **uni-nucleate? no** — it is the only one you control.
- **Smooth** = involuntary, non-striated, **uni-nucleate**, spindle-shaped; it controls the diameter of blood vessels and lines hollow organs; it has **no sarcomeres**.
- **Cardiac** = involuntary, **branching**, **intercalated discs**, striated with a **syncytium**-like branching network; it is the **myogenic** muscle (self-exciting); its involuntary + branching character is shared with smooth muscle in being involuntary.

- Common to cardiac and smooth: **involuntary**.
- **Myofibrils** are the contractile elements; the **sarcoplasm** is the muscle cytoplasm; the **sarcoplemma** is the muscle cell membrane; **endomysium** surrounds each muscle fibre.
- The **neuromuscular junction** is a **chemical synapse** between an **axon and the sarcoplemma**.
- **Shivering thermogenesis** = **involuntary muscle contraction**; the rib cage and pleura act passively during breathing, the diaphragm actively.

Sarcomere and the sliding-filament model

- The **sarcomere** = the unit between **two Z-lines** (one Z to the next); it is the repeating pattern of the myofibril. ×4
- Bands: **A band** = whole myosin; **I band** = actin only; the **H zone** sits at the **mid of the A band** and is the only myosin region.
- **Thick** filament (16 nm) = **myosin**; **thin** filament = **actin**, with **tropomyosin** and **troponin** bound to it. Each myosin is surrounded by **6 actin** filaments. ×3
- In a **relaxed** fibre: **tropomyosin blocks the myosin binding site** on actin, the **H zone is broad**, and no cross-bridges form.
- On stimulation: **Ca^{2+} leaves the sarcoplasmic reticulum** (that is its store), binds **troponin** — not actin, not tropomyosin, not the sarcoplemma — which pulls tropomyosin aside; cross-bridges form; actin **slides over myosin**; the **Z-lines move closer**, the **I band shortens** and the **H zone disappears**; the **A band stays the same** and the sarcomere shortens. ×8
- **ATP** is used for **cross-bridge breaking** (detachment of myosin), and energy is released when a **phosphate is removed** from ATP.
- On **recovery**, calcium is **pumped back into the sarcoplasmic reticulum**.
- **T-tubules** run perpendicular through the fibre and carry the action potential deep in, from the sarcoplemma into the sarcoplasmic reticulum; the SR's **Ca^{2+} gates** open as a result.
- **Myoglobin** is the muscle pigment that binds/stores oxygen; **haemoglobin** is the blood one. ×2
- Fibrous proteins do **not** carry oxygen in blood (that is haemoglobin's job); they do take part in clotting (fibrin), nail formation (keratin) and muscle contraction (actin/myosin).
- **Lactic acid** build-up → **muscle fatigue** (tiredness and pain), not cramp, tetany or paralysis; fatigue = **deficiency of glucose/ATP plus excess lactic acid**; **tetany** is an ionic imbalance (low Ca^{2+}).
- **Myosin is a globular protein** at its head but the filament is described as fibrous in some papers — actin, haemoglobin, myoglobin and insulin lack quaternary structure, myosin has it.

Other connective tissue and cytoskeleton

- **Cartilage** is a connective tissue that is avascular and heals slowly.
- **Intermediate filaments**: **8–10 nm**, made of **vimentin**, giving **mechanical support** (tubulin makes the 7 nm microtubules for cilia, flagella and cyclosis).
- **Actin** and **myosin** do not exhibit quaternary structure, unlike haemoglobin and myosin.
- All cells are genetically identical except **reproductive cells** (gametes).

Bled in from other units (same buckets)

- **Vestigial organs** = **ear muscles** + **vermiform appendix**; a remnant pelvic girdle in whales and snakes is also vestigial evidence. →u7

- The **autonomic nervous system** regulates visceral organs, glands and smooth muscles — it does **not** regulate skeletal muscles, and it does **not** control thoughts and emotions. →u5
- **Arthropods** = jointed legs + chitinous exoskeleton. →u7
- **Chemical digestion in the stomach** is the job of the middle (circular/longitudinal) muscle layers. →u14
- **Cartilage rings** stop the trachea collapsing. →u13
- The **myocardium** is the cardiac-muscle layer of the heart; the **papillary muscles** hold the tricuspid valve flaps. →u11
- The **sarcoplasmic reticulum** is the muscle's ER and its job is **calcium storage**; lipid metabolism is the **SER**'s job. →u4
- Comparative embryology shows **all vertebrates** are similar in early stages, and **gill pouches** appear in vertebrate embryos. →u7
- *Archaeopteryx* (1861, with teeth, a 20-vertebrae tail and clawed wings) shows birds descend from **reptiles**. →u7
- **B-lymphocytes** are named for the **bursa of Fabricius**; **passive artificial immunity** (antitoxin) fights active tetanus, rabies and snakebite; passive immunity is used against tetanus; **SCID** is treated by putting engineered cells into **bone marrow** (from the bone marrow itself comes "bone marrow" as the answer for neutrophils' release — they come from the bone marrow, are the most abundant WBC and are **not** made in the thymus). →u12
- Neutrophils come from the **bone marrow** and are the most abundant WBCs. →u12
- Marquetry of physics/chemistry questions misfiled here: torque = **(OP)(F sinθ)**; doubling all force parameters at θ = 90° makes the magnetic force **eight times**; **MRI** images **tumours and inflamed tissue**; bones look lighter on an X-ray because they **absorb more X-rays** (higher density); a 900 kg car going 5 → 30 m/s in 12 s feels an average force of **1875 N**; a force of **(OP)(F sinθ)**; **70–85% of CO_2** travels as **bicarbonate**; antibiotics come from **actinomycetes** and some bacteria; **candidiasis** is the thrush fungus. →u11, →u12, →u13
- **Proteins** establish the bone and cartilage matrix (**collagen**); **myosin** is described as a **globular** protein in one paper and a **fibrous** one in another — the myosin *filament* is thick and fibrous, its head globular.
- The **sarcoplemma** folds inwards to form **transverse (T) tubules**; the **sarcoplasmic reticulum** is the separate Ca^{2+} store.
- **Self-fertilisation** through successive generations leads to **hybrid breeds** — a genetics question. →u10
- The **H zone** in a relaxed sarcomere sits **in the mid of the A band**; on contraction the **I bands shorten** and the Z-lines draw together.
- **The hippocampus** is for **memory storage** (misfiled here). →u5
- **Osteoblasts** form bone; **osteocytes** maintain it; **osteoclasts** resorb it.
- The **elbow** is a **hinge joint** with a synovial cavity.
- **Osteoarthritis** is the most common arthritis.
- The **pubic symphysis** is a slightly movable **cartilaginous** joint of **fibrocartilage**.
- In relaxed muscle, **tropomyosin** (not actin, not troponin) blocks myosin from binding.
- Capillaries have the lowest blood velocity because they have the **largest total cross-sectional area**. →u11

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u10 — Inheritance (Mendel, dominance types, sex-linked, chromosomal)

Terminology

- **Homozygous** = both alleles of a gene pair the same (a true-breeding plant that self-fertilises into all-offspring-of-its-own-phenotype has a **homozygous** genotype). ×3
- **Genotype** = the genetic makeup inherited from the parents (your hair-colour makeup); **phenotype** = the observable physical characteristics. ×2
- **Karyotype** = the particular array of chromosomes an individual possesses.
- **Allele** occupies the **same locus on the respective homologues**; dominance is one allele's physiological effect over its partner at that locus.
- Mendel called his particulate factors **Elementen**; he worked with the **garden pea** (*Pisum sativum*), selecting **seven pairs** of contrasting characters.
- Mendel's second step after choosing pure-breeding tall and dwarf plants was to **cross them to produce hybrids**.
- **Test cross** = unknown × **tt** (homozygous recessive).
- A **hybridisation** = fusion of two genetically dissimilar parents by artificial pollination.
- **Apoptosis** = programmed cell death; **necrosis** = cell death from tissue damage.

Mendel's laws and ratios

- Monohybrid F_2 **genotypic** ratio = **1 : 2 : 1** → shows **segregation of alleles**; **phenotypic** ratio = **3 : 1** (e.g. 3 round : 1 wrinkled). ×3
- Dihybrid F_2 phenotypic ratio = **9 : 3 : 3 : 1**, which reflects **independent assortment**; the shuffling of alleles happens during **gamete formation**. ×2
- True-breeding round peas in F_2 = **25%**; a 1:1 ratio comes from **Yy × yy**.
- A 45 tall : 14 dwarf offspring means the parents were **Tt × Tt**.
- A heterozygous **AABbCc** plant makes **4** different kinds of gametes.
- Mendel's own RR × rr cross: monohybrid, each parent makes **only one kind** of gamete, F_2 shows **3 genotypes** but **2 phenotypes** — the false claim is "three phenotypes in F_2".
- The **law of segregation** says one individual out of four in F_2 has a phenotype that failed to show in F_1.
- When F_1 shows only the dominant parent's phenotype, the recessive allele is **dominant? no** — the trait is **dominant**, and the F_1 looks like the **dominant parent**.
- **Gene linkage** is the exception to Mendel's laws, and it is a **physical relation between genes** on the same chromosome. **Linkage** also distorts the 9:3:3:1 ratio.
- **Morgan** first discovered gene linkage in *Drosophila*, and his experiments showed deviation from independent assortment due to **linkage**.
- **Crossing over** is the exchange of segments between **homologous** chromosomes and is the phenomenon that increases variation; a higher recombination frequency means **more variation**.

Types of dominance

- **Complete dominance**: one allele fully masks the other.
- **Codominance**: both alleles expressed completely and fully → **AB blood group** is the classic example; *IAIA and IBIB are co-dominant* statements key as correct in ABO.
- **Incomplete dominance**: the heterozygote is **intermediate** — here the **genotypic and phenotypic ratios are the same** in F_1. ×6

- **Over dominance**: the **heterozygote exceeds** both homozygotes (Drosophila w/w^+ has a brighter eye pigment than w^+/w^+ or w/w). ×6
- **Pleiotropy** = one gene affects **two or more** (unrelated) traits.
- **Epistasis** = a gene at one locus suppresses or modifies the effect of a gene at another locus; the suppressed gene is **hypostatic**.
- **Polygenic inheritance** = several gene pairs at different loci acting **additively** on one trait — the basis of **continuous variation** (human skin colour is the example). ×2
- **Multiple alleles** = the **ABO blood group system**. ×3
- **Jumping genes** are the ones with **no fixed locus**.
- **Pattern baldness** is due to **hormonal differences** between sexes, and a **homozygous female is normal**; it is sex-influenced, not sex-linked.
- Two allelic genes at the same locus both fully expressed = **codominance**.

Sex-linked inheritance

- **Hemophilia** is **X-linked recessive**; **haemophilia A** = disturbed **Factor VIII**, **haemophilia B** = disturbed **Factor IX**. **Haemophilia A** is the more common one. ×2
- A hemophiliac male + carrier female → **50%** of sons affected (all daughters carriers); a carrier female × affected male → **50%** of sons; carrier × normal man → **half her sons** affected; normal man × heterozygous woman → **0%** chance for their **daughter** being hemophiliac (she would need both alleles).
- Hemophilia "zigzags" from the **maternal** grandfather, through a **carrier daughter**, to the **grandson** — never straight from father to son. ×6
- An **X-linked recessive** trait never passes father → son; it is equally passed to sons and daughters only if it is **autosomal**. ×2
- **X-linked dominant** (e.g. **hypophosphataemic rickets**, vitamin-D-resistant rickets): **all female offspring of an affected father are affected**, and an affected mother has a **50%** chance of passing it to any child.
- **Y-linked (holandric)** traits: gene on the **Y**, always **more common in males**, passed from an affected father **only to his son**.
- Sex-linked recessive traits are **more likely to affect men than women** — colour blindness, haemophilia.
- **Colour blindness** = **deuteranopia** in the papers' naming; red colour blindness means red and green both look **grey** — the *drosophila* "eyed" trait is sex-linked too (white-eyed female × red-eyed male gives a **0.5** chance of colour-blind male offspring in the keyed version).
- **Hypophosphataemic rickets** is the X-linked **dominant** disorder; **haemophilia, colour blindness, Becker muscular dystrophy** are X-linked recessive.
- **Drosophila bobbed** gene is **X-linked**.

Blood groups

- **ABO** = **multiple alleles** (IA, IB, i) on an **autosomal** chromosome; IA and IB are **codominant**; it **codes for antigens** (the false claim is "doesn't code for any antigen").
- A person with **group O** whose children are AB cannot have a mother **homozygous for IA** (the other options can work); group O is ii.
- A **baby girl with haemophilia** must have a **carrier (or affected) mother and a haemophiliac father**.

Mutations

- A change in a DNA molecule's structure making a **different allele** = **gene (point) mutation**.

- **Phenylketonuria** = a **point mutation**; **sickle-cell anaemia** = **base substitution** (glutamic → valine).
- **Chromosome inversion** and other structural changes are **chromosomal** mutations.
- **Mutation** is the **ultimate source of new genetic variation**; **crossing over** and **independent assortment** also create variation, but **natural selection** does not create new variation — it sorts it.
- A **selectively advantageous** mutation increases in frequency and **spreads through the population over generations**; a **bottleneck effect** is the population shrink after a natural disaster, and the **founder effect** is from a small colonising group.

Chromosomal abnormalities

- **Turner's syndrome = 44 + XO** — sterile female, infertility, short stature, webbed neck, low hairline. ×4
- **Klinefelter's = 44 + XXY** — enlarged breasts and small testes in males, a sterile male in *Drosophila*. ×4
- **XXY is sterile female in humans but sterile male in Drosophila** — the sex-chromosome count behaves differently across species. ×8
- **Down's syndrome = trisomy 21 (autosomal non-disjunction)** — mental retardation, short stature, broad face, squint eyes. ×2
- **Edward's = trisomy 18**; **Patau's = trisomy 13**; **Jacob's** is a sex-chromosome abnormality.
- **Alzheimer's disease** is the age-related one.

Bled in from other units (same buckets)

- **Evolution** = change in allele frequencies in populations over generations; the struggle for existence is from **overproduction**; the best adapted leave the most offspring. →u7
- **14 chromosomes** are found in the **garden pea**.
- The **tRNAVal** is the tRNA that inserts the valine of the sickle-cell haemoglobin. →u3
- **Self-fertilisation** in successive generations leads to reduced variation and inbreeding effects.
- **Chromosomal abnormalities of Turner = XO, Klinefelter = XXY** also surface in u4 and u7.
- **Down's syndrome is autosomal non-disjunction**; **Jacob's** is the sex-chromosome form.
- Haemophilia patients can bleed to death from small cuts and bruises.
- **Crossing over** increases the chance of variation; **linkage** does not.

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u11 — Circulation (heart, vessels, blood, lymph)

Heart chambers and valves

- The **right atrium** receives **deoxygenated blood** from the superior and inferior vena cava; the **left atrium** receives oxygenated blood from the **four pulmonary veins**.
- The **aorta** originates from the **left ventricle**.
- **Tricuspid (right AV) valve** separates the **right atrium from the right ventricle**; the **bicuspid/mitral** valve goes from **left atrium to left ventricle**.
- **Semilunar valves** sit at the aorta and pulmonary artery outlets and **prevent backward flow** into the ventricles — they are **not** in veins.
- A human heart has **6 valves** in total (2 atrioventricular + 4 semilunar halves), though most papers count **3 valve groups**.

- **Chordae tendineae** are the fibrous cords in the **ventricles** attached to the **tricuspid valve** and papillary muscles; weakening the papillary muscles lets blood back-flow into the **right atrium**. ×2
- The heart wall layers: **endocardium** (inner, and the layer the **valves** are formed from), **myocardium** (the **cardiac muscle** wall — the only cardiac-muscle layer), **epicardium** (outer). The closed sac around the whole heart is the **pericardium**.
- The **SA (sinoatrial) node** sits in the wall of the **right atrium**, upper dorsal, and is the **pacemaker** that normally regulates the heartbeat; impulses go **SA node → AV node → Bundle of His → Purkinje fibres**.
- The delay between SA and AV node conduction is about **0.10 s** (the papers' keyed value).
- The ventricle's thick-walled chambers sit on the **apex / lower side** of the heart.

Cardiac cycle

- One complete heartbeat = **1 systole + 1 diastole**, lasting about **0.8 s**.
- **Ventricular systole** closes the atrioventricular (bicuspid/tricuspid) valves and opens the semilunar ones; ventricular diastole opens the AV valves again.
- The **LUB** sound = AV valves closing at the start of ventricular systole; the **DUB** = semilunar valves closing as the ventricles relax. Heart sounds come from **valve closure**, not contraction.
- **Diastole** = the chambers **relax and fill with blood**.
- The **cardiac output** = the volume of blood the left ventricle pumps into the systemic circuit per minute.
- Cardiac muscle is the muscle of the **heart walls only** (not the valves, not the aorta).

Blood vessels

- **Veins** = **larger bore, thin walls, no pulse**, and **valves present** (except in the upper part of the body — that exception is the false statement); wall ~two cells thick; diameter 40–50 µm; flow is **toward the heart**.
- **Arteries** = **thick, muscular walls**, no valves, pulsatile.
- **Capillaries** = walls of **one cell**, walls are **highly permeable**, and they have the **lowest blood velocity because they have the greatest total cross-sectional area**.
- Microcirculation order: **arterioles → metarterioles → thoroughfare channels → capillaries**.
- The **descending aorta** bifurcates into the **iliac arteries**, whose femoral branches feed the thigh muscles.
- The **pulmonary vein** carries **oxygenated** blood (the one vein that does); the **pulmonary artery** carries deoxygenated.
- A **four-chambered heart with a right systemic arch only** = **birds** (mammals have the left systemic arch).
- **Bernoulli's principle**: blood pressure drops when velocity **increases in a narrow vessel**.

Blood

- **Fibrinogen** is the plasma protein involved in **clotting**; **prothrombin** also takes part; **albumin** maintains **osmotic balance**.
- **Blood group antigens** are **glycoproteins** (and glycolipids); they sit on the RBC surface.
- Haemoglobin transports O_2 and CO_2 and **buffers blood pH** — it does **not** regulate blood pressure in stress.
- **Carboxyhaemoglobin (10–20%)** forms when **CO_2** combines with the **haem portion** of haemoglobin.
- **Leukaemia** = an increased number of **blood cells** (white cells) in the blood.
- **Hepatitis B**'s major route is **blood transfusion** (and transfusion/blood contact generally).
- **Cholesterol** deposits plaque-like masses in arteries, causing a **heart attack**.

- Blood enters the **glomerulus through the afferent arteriole**; Bowman's capsule continues into the **proximal convoluted tubule**.
- **Anti-A and anti-B antibodies** in serum = **blood group O** (the only group with both); both antigens present = **group AB**; group A− can be donated to **A positive**.
- **O−ve** is the **universal donor**; **AB+ve** is the universal recipient. ABO gives **4 phenotypes**; the ABO system was discovered by **Landsteiner**; the Rh factor was named after the **Rhesus monkey**, and Rh gives **2 phenotypes**.
- **IAi × IAi** gives **3/4 group A and 1/4 group O** children; an A father with a B child can be **IBi**; a group-O child from a group-B father means the father is **IBi**.
- Group A × group B parents can have a child of **any of A, B, AB and O**.
- **Anti-A and anti-B antibodies** appear **in the first few months of life**, not on first exposure.

Lymph

- Lymphatic tissue fluid = **lymph**.
- The **thoracic duct** is the largest lymph duct and drains into the **subclavian vein**; lymph empties into the bloodstream at the **subclavian vein**.
- Loss of the lymphatic system compromises **blood composition and defence**, not oxygen-carrying capacity.
- **Lymphatic blockage** first causes **excess fluid accumulation in the tissue** (oedema).

Homeostasis of the blood

- **High osmotic pressure of blood** (dehydration) triggers **ADH** secretion.
- **Alcohol, caffeine and atrial natriuretic factor** all favour large quantities of dilute urine — **renin** does not.
- A **negative feedback** mechanism example = **control of body temperature** (platelet aggregation, labour and milk ejection are positive feedback).
- Vasodilators **relax smooth muscle** in vessels to **reduce blood pressure**.
- Sympathetic over-activity raises **heart rate** and **dilates the pupils**; the parasympathetic constricts bronchi and promotes digestion.
- The **autonomic nervous system** does **not** control **limb movement** (that is somatic).

Bled in from other units (same buckets)

- **ABO = multiple alleles**, symbol **I**, on an autosomal chromosome; the character determined by three alleles = human blood group. →u10
- **Polygenic** = human skin colour and eye colour; **Rh** is a single-gene system. →u10
- **Nissl's granules** are in **nerve cells** (not WBCs, RBCs or platelets). →u5
- The **knee-jerk** reflex is the exception to polysynaptic transmission (it is monosynaptic); heartbeat, breathing and playing games are polysynaptic. →u5
- **Histamine** is produced by **basophils**; **E. coli**-style bacteria destroy red corpuscles and cause pernicious anaemia. →u12
- **Necrosis** vs **apoptosis** belongs to u10's cell-death items.
- English items misfiled here: "it is just a **cyclone** in the teacup"; "**resuscitate** him".
- Capillary rise: the tube of **smaller radius shows a higher rise** of water.

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u12 — Immunity

The four kinds of immunity

- **Natural active** = you catch the disease and survive, then develop your own immunity (exposure to HIV and recovering → natural active). ×6
- **Natural passive** = antibodies handed to a baby in **colostrum / mother's milk**.
- **Artificial active** = a **vaccine** (attenuated culture of bacteria, live or inactivated polio) — slow to build, long lasting.
- **Artificial passive** = injecting **antibodies / antiserum / anti-venom / immunoglobulins / plasma** from a recovered patient — **immediate** protection, short lived.
- **Passive** immunity is what gives **immediate** protection; injecting antibodies is **passive**, and the snake-bite antivenom is **passive** too.
- Recovered-COVID plasma works because of the **antibodies** in it, not interleukins or interferons.

The lymphocytes

- **B lymphocytes** are made and **matured in the bone marrow** (named for the **bursa of Fabricius** in birds). They **secrete antibodies** — plasma cells come from B cells in the **humoral** response. ×3
- **T lymphocytes mature in the thymus**; the thymus is the gland of T-cell maturation and it produces lymphocytes. ×2
- **Cell-mediated immunity** = T cells recognise the antigen and attack the microorganism or a transplanted organ; **humoral immunity** = B cells and their antibodies. ×4
- **Helper (T4/CD4) T cells** are required to activate B cells; **cytotoxic T** and **memory T** complete the cell-mediated arm.
- Antibodies are made by **B lymphocytes** — not T cells, not RBCs, not platelets. ×2
- **Lymphocytes** (with monocytes/macrophages) are the cells central to immunity; **thrombocytes (platelets)** are not.

Antibody structure

- An antibody has **2 heavy + 2 light chains**, 4 polypeptide chains total, joined by **disulphide bridges**; the **variable region** is at the **upper** end and is what **recognises the antigen**; the **constant region** is at the lower end. ×3
- Antibodies **neutralise** antigens, **precipitate** them and **enhance phagocytosis** — they do **not** secrete cytokines (that is a living cell's job).
- **Monoclonal antibodies** in diagnosis **detect specific antigens** in clinical samples.

Innate / non-specific responses

- Non-specific barriers: **mucus, HCl, spermine** (in semen), phagocytes, inflammation, fever.
- **T cells** are specific, not non-specific — they are the exception in the "which is not non-specific" question.
- The **inflammatory** chemical of the upper respiratory tract is **histamine**, produced by **basophils** (and mast cells); histamine makes blood vessels **dilate**. ×6
- **Bradykinin** is produced at the site of tissue injury and promotes fluid leakage; **interferons** are the cytokines that **activate natural killer cells** and control viral infection — they do not "secrete interleukin". ×2
- The **complement** and inflammatory cascade causes the redness and swelling of inflammation.

- **Neutrophils** are the **most abundant WBC** and the first to arrive at an infection; **monocytes** become tissue **macrophages**; **macrophages** and neutrophils are the main phagocytes. ×2
- **Granulocytes** = **neutrophils, eosinophils and basophils** (monocytes and macrophages are agranulocytes).
- Phagocytosis is what lysosomes and macrophages/kupffer cells do.
- The **complement** system opsonises pathogens; **opsonins** mark them for phagocytosis.

Lymphoid organs and lymph flow

- **Lymph nodes** filter lymph; **lymph vessels** move it. The **spleen** is a lymphatic organ (it also filters blood and removes old red cells); the **thymus**, **tonsils** and **Peyer's patches** are lymphoid too.
- Lymph flows **toward the thoracic lymph duct** and empties into the **subclavian vein**; the lymph's **final destination** is the subclavian vein.
- Lymph through the lymphatic vessels = **lymph**; the vessel in the intestinal villi is the **lacteal**, which carries absorbed fat.
- Swollen lymph nodes after a throat infection = an **active immune response**.
- **Lymph nodes vs vessels**: vessels move lymph, nodes filter it.

Blood transfusion rules

- Transfusion matches **the donor's antigen against the recipient's antibodies** — donor antigen must not be attacked by the recipient's antibodies.
- **Rh^+ blood** carries **Rh antigens on the RBCs** (the Rhesus **monkey** is where the factor was first studied).
- **Rh antibodies** in Rh-negative people appear only **after exposure to the Rh antigen** — never before birth, never naturally after birth.

Chemotherapeutics, antisepsis and sterilisation

- **Chemotherapeutics** = chemicals used to kill or inhibit microbes **in living tissue**. ×6
- **Disinfectants** work on **inanimate** surfaces; **antiseptics** on living tissue; **antibiotics** are chemicals made by microorganisms that destroy other microbes (from **actinomycetes** and some bacteria).
- **Pasteurisation** preserves **milk products**; **vaccination** uses an **attenuated** culture.
- The biggest problem with antibiotic misuse = **increased resistance in pathogens**.
- **BCG vaccine** is used against **pulmonary tuberculosis**.

Diseases

- **HIV** targets **T-helper (CD4) lymphocytes** — both B and C options describe the same cells.
- **SCID** = missing **adenosine deaminase**; treatable by putting genetically engineered cells into **bone marrow**.
- **Cystic fibrosis** affects **endothelial / epithelial** cells (the sweat and mucus glands); liposome gene therapy coats the healthy gene.
- **Arthritis** = inflammation of a joint, an **autoimmune** disease and the leading cause of disability over 65 — it is *not* inflammation of a nerve.
- **Leukaemia** = too many white blood cells.

Bled in from other units (same buckets)

- **Thymus** produces lymphocytes; the **spleen** filters blood and removes old red cells. →u11
- **Monoclonal antibody** technology and the engineered **malaria vaccine** belong to u16 biotechnology. →u16
- **Interferons** control viral infection; **bradykinin** and **histamine** drive inflammation.

- **Kupffer cells** of the liver are the resident phagocytes.
- **Cerebral cortex** is the outer grey-matter layer (a CNS item misfiled here). →u5
- The waste excreted by a healthy human is **urea** (not ammonia, uric acid or creatinine). →u15
- Swollen lymph nodes = active immune response; lymph flows to the **subclavian vein** via the thoracic duct. →u11

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u13 — Respiration (human)

Airway anatomy

- The **pharynx** is the part common to both the digestive and the respiratory systems, and its main respiratory role is being the **passage for both food and air**. ×2
- The **epiglottis** is the flap that stops the bolus entering the trachea; it is also what closes the nasal opening during swallowing (along with the **soft palate**, which pushes up). ×3
- **Swallowing** is initiated by the **tongue**, which forces the epiglottis into a horizontal position and moves the soft palate up.
- The **trachea** lies **ventral** (in front of) the oesophagus; the **pharynx** sits above the larynx.
- **Cartilage rings** in the trachea stop it collapsing and keep the air passage open.
- **Ciliated epithelium + goblet cells** line the **trachea and bronchi** — **alveoli have no cilia**.
- **Smooth muscles** are in the trachea, bronchi and bronchioles — **alveoli have none**.
- **Bronchioles** end in the **alveoli and alveolar sacs**; alveoli are the actual **gas-exchange surface**, and each air sac is made of several microscopic **single-layered** alveoli.
- Alveoli are elastic because of **collagen and elastin** — these fibres expand and recoil.
- The lungs are enclosed in the double-membraned **pleura** (two layers); the area between the two lungs is the **mediastinum**; the **hilum** is the lung's root.
- The lungs are closed sacs open to the outside only via the **trachea, nostrils and mouth** — never the oesophagus.
- The **diaphragm** separates the thoracic from the abdominal cavity; the stomach sits below it on the **left side of the abdomen**.
- The **diaphragm** is the main muscle of breathing, while the **rib cage and pleura act passively**.
- Frog breathing: fresh air enters the bucco-pharyngeal cavity through the nostrils and old air is held under pressure in the **glottis**.

Mechanics and volumes

- On **inhalation** the diaphragm moves **down** and the ribs move **up and out**, so the **chest cavity volume increases and the lung pressure decreases**. ×6
- **Tidal volume** at rest ≈ **0.5 L**; **residual volume** (left after forced expiration) ≈ **0.5–1.5 L**; **total lung capacity** ≈ **5–6 L** (total inside capacity ~5 L). ×2
- The **residual volume** is the air that always remains in the lungs.
- Lung capacities of interest: tidal 0.5 L, expiratory reserve ~1.1 L, inspiratory reserve ~3.1 L, residual ~1.2 L, total ≈ 5 L.

Gas transport

- About **70%** of CO_2 is carried as **bicarbonate** ions; the **chloride shift (Hamburger's phenomenon)** moves Cl^- into the RBC while HCO_3^- leaves. ×2
- **Formation of oxyhaemoglobin (HHb)** is *not* part of tissue-level gas exchange — that happens in the lungs; at the tissues you get **HHb formation, sodium bicarbonate and the chloride shift**.
- **Carboxyhaemoglobin** forms when **CO** binds the **iron** part of the haem (10–20% saturates).
- The **pulmonary vein** returns oxygenated blood; the **pulmonary artery** carries deoxygenated.
- The **carbonic anhydrase** in RBCs speeds $CO_2 + H_2O \rightarrow H_2CO_3 \rightarrow H^+ + HCO_3^-$.

Diseases and disorders

- **Respiratory distress syndrome** in premature babies = **deficiency of surfactant** (surfactant is a **lipoprotein**). ×2
- **Tuberculosis** is a respiratory disease (and in one paper keyed as the STD among the options — the STD is **syphilis** in the same question set).
- **Lung cancer** is the common respiratory malignancy; **testicular cancer** is the young-men's high-cure-rate one.
- The **larynx's** primary function is **voice production** (not humidifying, filtering or gaseous exchange).
- **Respiratory surface** must be thin, moist and have a large area with a good **blood** supply.
- Excretory organs include the kidney, skin and lungs — the **appendix** is not excretory.
- **Pseudomonas** is a **spiralled** bacterium; **Staphylococcus pneumoniae**'s non-virulent strain is **non-capsulated R-type** (the virulent one is capsulated S-type, the Griffith transformation).
- **Nose and trachea** are both supported by **hyaline cartilage** (the epiglottis is elastic cartilage).

Bled in from other units (same buckets)

- **Surfactant** is a **lipoprotein** — a u3 conjugated-molecule fact. →u3
- **Hyaline cartilage** is the most abundant cartilage (u9). →u9
- **Diaphragm** as a smooth muscle, and the **intercostal muscles** of the chest wall, belong to u9. →u9
- **Hamburger's / chloride shift** is a u11 circulation fact. →u11
- **The spleen** filters blood and removes old red cells (u11). →u11
- **Urea** is the healthy human's excretory product (u15). →u15
- Chemistry that landed here: the geometry of **but-2-ene** is its cis/trans isomerism case, and the hydrogen needed to make 5 kg of ammonia is ~**15 g** worth of H_2 — an industrial-gas calculation. →u16
- **Digestion of protein starts in the stomach** (u14). →u14

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u14 — Digestion

Mouth and saliva

- Digestion **begins in the buccal cavity** and ends in the small intestine. ×2
- The **parotid glands** sit **behind the jaws** (the other two are submandibular/submaxillary and sublingual). ×8
- Mastication makes the sublingual and back-of-mouth glands secrete **saliva**; the ground food mass is the **bolus**.

- **Amylase (ptyalin/diastase)** digests **starch** in the mouth; digestion of **carbohydrate** is the only one that starts in the oral cavity. ×3
- **Lingual lipase** comes from **Ebner's gland**.

Oesophagus

- Food is forced down the oesophagus by **peristalsis** (swallowing/deglutition initiates it).
- **Pyrosis (heartburn)** results from inefficiency of the **oesophagus / cardia** — acid reflux.

Stomach

- The portion right after the oesophagus is the **cardiac portion**, then fundus, body, pyloric.
- Gastric glands have **3 (four)** cell types: **parietal (oxyntic) cells** make **HCl** *and* **intrinsic factor**; **chief (zymogen) cells** make **pepsinogen**; **mucus/goblet cells** make mucus; endocrine cells make **gastrin**. ×6
- **Protein breakdown** in the stomach is done by **chief cells** (via pepsin).
- **Gastrin** is secreted from the **pyloric region / antrum** of the stomach, and it **stimulates gastric juice**.
- **Pepsinogen** → **pepsin** by **hydrochloric acid** (low pH).
- **Intrinsic factor** (from parietal cells, i.e. the **stomach**) is needed for B_{12} uptake.
- Stomach wall muscles churn food with gastric juice into **chyme**.
- Acid reaching the duodenum is the **acid in food** that stimulates secretin.

Small intestine

- **Digestion is completed in the small intestine**; it is the absorptive organ too.
- **Secretin** is released into the blood from the **duodenal mucosa** by acid; it stimulates pancreatic juice (the bicarbonate part).
- **Cholecystokinin (CCK) releases bile from the gall bladder**.
- **Enterokinase (enteropeptidase)** is made by the **duodenal lining** and activates **trypsinogen** → **trypsin**. ×3
- **Chymotrypsinogen** is a component of **pancreatic juice**; it is also activated as a zymogen.
- Proteins after digestion are absorbed as **amino acids and small polypeptides**.
- The **ileocolic sphincter** at the end of the ileum lets a little residue into the large intestine. ×6
- The **pyloric sphincter** lets chyme from stomach → small intestine; the **cardiac sphincter** guards the other end.
- Lipids: **mechanical** digestion = the stomach (churning) / bile emulsification, **chemical** = **pancreatic lipase**.
- The carbohydrate-digesting enzyme in pancreatic juice is **amylopsin (pancreatic amylase)** — not ptyalin (that is salivary).
- **Pancreatic lipase** works at alkaline pH; **pepsin** is the acidic one.
- **Lactase** is the odd enzyme out among pepsin, trypsin and chymotrypsin (it is a carbohydrase, not a protease).

Pancreas, liver and bile

- **Bile** = a **watery, enzyme-free** secretion containing the **breakdown products of haemoglobin** (bile pigments), **bile salts** and **cholesterol** — **lipase is not** in bile. ×2
- Bile is made in the **liver**, stored in the **gall bladder**, and poured into the duodenum through the **common bile duct**.
- **Emulsification of fats** is bile's job.
- **Jaundice** = elevated **bile pigments in the blood**.

- Removing the gall bladder causes **uncontrolled release of bile** → poor fat digestion.
- **Secretin** stimulates pancreatic secretion; the hormone that *inhibits* it in the papers' keying is **gastrin** (the intended contrast is that gastrin acts on the stomach, not the pancreas).
- **Gastrin + secretin** together drive the pancreas in the historical papers' version.
- **Hepatitis = inflammation of the liver**; **hepatitis A** spreads by the **faecal-oral** route, **hepatitis B/C** are blood-borne (hepatitis B also passes **mother → child during pregnancy**), and **hepatitis D** is the **viroid** disease.
- **Mesenteric veins** drain the **large intestine**.

Large intestine

- Its jobs: **absorb water**, **absorb electrolytes**, and **make vitamins** (large-intestine microflora make **vitamin K**, with symbiotic bacteria). ×2
- It does **not** absorb amino acids (that is the small intestine).
- Parts: caecum, colon, sigmoid colon, rectum — the **appendix attaches to the caecum**.
- **Villi are not** in the large intestine (they are small-intestine).

Goblet cells and mucus

- **Goblet cells secrete mucus** — never HCl, enzymes or hormones. ×5

Enzyme zymogens

- Secreted **inactive**: pepsinogen, trypsinogen, chymotrypsinogen (and erypsinogen); the one secreted **active** is **amylase** (also lipase in some lists). ×3
- All proteolytic enzymes are released as **zymogens** except **pepsin** itself is active once formed by HCl.

Absorption

- **Glycogen** is stored in the **muscle and liver**. ×2
- The large intestine absorbs **water and salts**; the **lacteals** carry absorbed **fat**.
- **Chylomicrons** = fat + protein, formed in the gut wall.

Feeding disorders

- **Anorexia nervosa** = stopping eating out of fear of obesity (the 18-year-old with underdeveloped feminine characteristics and psychological immaturity). ×2
- **Bulimia nervosa** = binge-then-purge; **orthorexia** = obsession with "healthy" food; **dyspepsia** = indigestion.

Parasites and worms

- **Fasciola** = **liver fluke**, an endoparasite of sheep in the **liver and bile duct**; its primary host is the **sheep**, with the **snail** as intermediate host.
- **Schistosoma** lives in the host's **blood**.
- **Taenia** = tapeworm; **Ascaris** = roundworm (endoparasite); **leech** = the **ectoparasite**.
- **Dugesia** is the **free-living carnivorous flatworm**.
- *Coelenterate* colony members that do different jobs are **zooids**.
- **Alternation of generation** = sexual form producing asexual form and vice versa.
- **Vermiform appendix** is the human **vestigial organ** here.

Bled in from other units (same buckets)

- **Cyclic photophosphorylation** does *not* include **photolysis of water** — that is the non-cyclic path. →u2
- **Storage (lysosomal storage) diseases** come from defective **lysosomal enzymes** — Tay-Sachs and the like. →u4, →u6
- **Symbiotic bacteria** in the large intestine make **vitamin K**. →u7
- **Glycogen storage = liver and muscle** is a u3 polysaccharide fact. →u3
- **Aldosterone** conserves **sodium**; **maximum reabsorption** is in the **proximal tubule**. →u15
- **Ductless glands = endocrine glands**. →u5
- **Anorexia nervosa** (fear of obesity) vs **bulimia nervosa** (binge + purge).
- **Hepatitis is the blood-borne disease** among cholera, influenza and candidiasis.
- **Peroxyacetyl nitrate** irritates the **eyes** (air pollution). →u13
- **Redox = electron transfer**; **blood movement in veins is not capillary action**.
- **The appendix attaches to the caecum**.
- **Liver failure** can cause acute kidney failure (hepatorenal syndrome). →u15

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u15 — Homeostasis (kidney, osmoregulation, excretion)

Definition

- **Homeostasis** = maintaining the internal environment against fluctuations of the external one. Thermoregulation is one kind of it; evolution is the accumulation of adaptations over time.

Nephron — filtration and reabsorption

- Filtration happens at the **glomerulus + Bowman's capsule**.
- **Most reabsorption** happens in the **proximal tubule** (the useful constituents too); this is also where water and sodium are reclaimed, along with the collecting duct. ×3
- The **countercurrent multiplier** is the **loop of Henle** (in the renal medulla); water is reclaimed by that mechanism in the **collecting duct**.
- The **ascending limb of the loop of Henle** actively pumps out **Na^+**, promoted by **aldosterone**, and it does **not** let water out. ×2
- **Hypertonic (concentrated) urine** is made by the **loop of Henle**, chiefly via **juxtamedullary nephrons**, which sit at the cortex–medulla border and handle the water-shortage case. ×2
- **ADH** acts on the **distal tubule and collecting ducts**, changing their permeability so water is reclaimed. ×3
- Under **ADH**, the kidney makes **hypertonic urine with decreased volume** — the keyed combination. ×6

Anatomy of the kidney

- The **renal pelvis** collects urine before it enters the **ureter** (cortex and medulla do not).
- Urine → renal pelvis → ureter → urinary bladder → urethra.
- The **renal cortex** holds cortical nephrons; the **medulla** holds the juxtamedullary ones.

Hormonal control

- **ADH / antidiuretic hormone** → water reabsorption → concentrated, low-volume urine; it rises when blood water content falls or blood osmotic pressure is high (the dehydrated desert mammal).

- **Aldosterone** → Na^+ reabsorption in the nephron; conserves sodium.
- **Parathyroid hormone** acts on the kidney for calcium; **atrial natriuretic peptide** opposes it and promotes dilute urine.
- The **thirst** and the **homeostatic thermostat** both sit in the **hypothalamus**.

Waste and excretion

- The main nitrogenous waste of humans = **urea**. The most **toxic** nitrogenous excretory product = **ammonia**. ×2
- Humans are **ureotelic**; freshwater fish and many aquatic animals are **ammonotelic**; birds and reptiles are **uricotelic**.
- **Excretion** = removal of metabolic waste from the blood; **osmoregulation** is the water-salt balance; **renal failure** is the backup of urea.
- **Kidney stones**: the best preventive measure is **drinking plenty of water**.
- **Acute kidney failure** in the keyed set is caused by **liver failure**.
- **Peritoneal dialysis** introduces the dialysing fluid into the **abdomen** (peritoneal cavity); the dialysing fluid matches blood osmotic pressure but **lacks the nitrogenous waste**.

Other animals' excretory organs

- **Arthropods** = **Malpighian tubules**.
- **Planaria** = **protonephridia** (flame cells).
- **Annelids and cephalopods** = **metanephridia**.
- **Vertebrates** = **nephrons / kidney**.

Plant water transport (same buckets)

- The **Casparian strip** is in the **endodermis** and makes the **apoplast pathway discontinuous** there, forcing water into cells. ×6
- **Transpiration pull with cohesion** is the most accepted explanation for the ascent of sap; root pressure, capillarity and atmospheric pressure are the older alternatives.
- **Water rises in a capillary** when the **adhesive force exceeds the cohesive force**.
- Water is a universal solvent because of its **high polarity**; it sticks to polar wood because of **adhesion** (cohesion is water-to-water). ×2

Thermoregulation

- **Frostbite** is due to **vasoconstriction** in the cold.
- **Vasodilation + sweating** cool the body; **shivering + vasoconstriction** warm it.
- Body temperature is a **negative feedback** loop; **labour contractions**, platelet aggregation and milk ejection are **positive feedback**.

Bled in from other units (same buckets)

- The **encapsulated** bacterium survives better because of high pathogenicity, dehydration resistance and adhesion — **easy recognition by the host** is the exception. →u12
- Plants with **aerenchyma** (air tissue) are **hydrophytes**. →u4
- **Urea** is the most widely used nitrogen fertiliser in Pakistan, formula **NH_2CONH_2**; **Wöhler** first synthesised it from **ammonium cyanate**. →u3

- Chemistry that landed here: **melting of ice** is the endothermic process, and in an endothermic reaction the **products** have more heat content than the reactants; **PVA** is the colourless non-toxic resin used as an adhesive and as a binder for **compact discs**; **casein glue** is the natural adhesive; **urea urea = N_2H_4CO**; **radon** is an **α-emitter** used in **cancer** radiotherapy; **peroxyacetyl nitrate** irritates the **eyes**. →u3, →u13
- English items misfiled here: youngsters who indulge in love affairs are usually **addled** in worldly manners.
- **ADH** comes from the posterior pituitary, made in the hypothalamus; **aldosterone** from the adrenal cortex. →u5
- **Diabetes insipidus** = ADH deficiency; **diabetes mellitus** = insulin deficiency. →u5
- **Atrial natriuretic peptide**, alcohol and caffeine all give large quantities of dilute urine; **renin** does not. →u11
- The **blood osmotic pressure** rising triggers ADH — the desert-mammal dehydration case. →u11
- The **spleen** filters blood and removes old red cells. →u11
- Uric-acid crystal deposition in joints = **gout**. →u9
- **Vestigial** organs such as the vermiform appendix are an evolution item. →u7
- The **renal artery** → **afferent arteriole** → **glomerulus** → **efferent arteriole** → **peritubular capillaries** loop is the kidney's blood pathway. →u11
- The **ureter** carries urine to the bladder; the **urethra** is the outlet.
- **Urea** is made in the liver from amino acids and excreted by the kidney.
- **Sweat glands** are exocrine (duct) glands; the **thirst centre** is in the hypothalamus.
- The kidney also buffers blood pH by excreting H^+ and reclaiming bicarbonate.

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u16 — Biotechnology

Core idea and cloning

- **Biotechnology** = using living organisms in industry to make useful products.
- **Cloning** = producing genetically identical copies of an organism by asexual reproduction; the identical individuals are **clones**. ×2
- Plants carrying foreign DNA = **transgenic plants**. ×2
- **Genetic engineering** joins DNA from different sources; the result is **recombinant DNA (rDNA)**.
- The modified plasmid or phage DNA that carries the new gene is called **recombinant DNA / rDNA**.
- **DNA probes** are made of **radioactively labelled nucleotides**.

Tools and vectors

- **Enzymes** are the tools for manipulating DNA in recombinant DNA technology.
- **Restriction endonucleases** are the molecular scissors, cutting the **phosphodiester bond** of duplex DNA. ×3
- **DNA ligase** is the enzyme that joins two double-stranded DNA fragments.
- **Plasmids** are the common vectors in recombinant DNA technology; a **phage** vector, and the classic one is **lambda phage**. ×3
- Bacteria take up recombinant plasmids when treated with **$CaCl_2$**.
- **Agrobacterium tumefaciens** is the bacterium used to make transgenic plants such as **golden rice**.

PCR

- **PCR** is the process that makes multiple copies of the desired gene; its sequence is **heating** → **cooling** → **add primer** → **copying of the strand**. ×2
- **Taq polymerase** is the heat-stable polymerase; **heat** separates the DNA strands.

Reverse transcription and gene therapy

- DNA made from mRNA by reverse transcriptase = **complementary DNA (cDNA)**.
- **Gene therapy** for **cystic fibrosis** uses **liposomes (prosomes)** coated with the healthy gene; for **SCID (adenosine deaminase deficiency)** it works by putting genetically engineered cells into the **bone marrow**. ×2
- **Insulin** is produced for diabetics by **genetically engineered *E. coli***; growth hormone too. ×3
- **Monoclonal antibodies** are used in diagnosis to **detect specific antigens** in clinical samples; a **biotech malaria vaccine** works by creating a **targeted immune response** against the parasite.

DNA fingerprinting

- **DNA fingerprinting** is the technique used to identify criminals, and the **microsatellites** are what produce the distinctive pattern on the autoradiograph.
- Age of fossils is determined by **radioactive dating** (not DNA fingerprinting or PCR).

Plant tissue culture

- **F. C. Steward** used the **carrot** for his tissue-culture / totipotency experiment.

Biological control

- **Biological control** = destroying pests with the same living organisms or natural enemies (not pasteurisation, not integrated disease management, not genetic engineering).

Bled in from other units (same buckets)

- **Positive feedback** examples: **labour contractions** (body temperature, insulin production and thyroxine release are negative feedback). →u5
- **Genetic drift / gene pool / mutation** are evolution mechanisms, not continental drift. →u7
- **Genetic engineering cells** go into bone marrow for **SCID**; **prosom** liposomes for **cystic fibrosis**. →u12
- **Agrobacterium** and **E. coli** are the two workhorse organisms; **Mykobacterium** and *Xanthomonas* are the distractors.
- Physics that landed here: the area of a parallelogram with sides **A** and **B** is **AB sinθ**; if **2i + 3j + 8k ⊥ 4j − 4i + αk** then **α = −1**. →physics
- **The gene probe** is the radioactively labelled nucleotide tool.
- **Restriction enzymes** cut DNA; **ligase** joins it; **Taq polymerase** copies it; **lambda phage** carries it — the four-part toolkit.
